

APPLIED FINANCIAL ECONOMETRICS

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2025 - 2026

Plan

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2 High Frequency Data (H.F.D.)

3 Volatility Forecasting Using High-Frequency Data

- Realized Measures
- Forecasting Models

4 Market Risk Modelling

- General Metrics

■ Value-at-Risk

■ Expected Shortfall

■ Backtesting

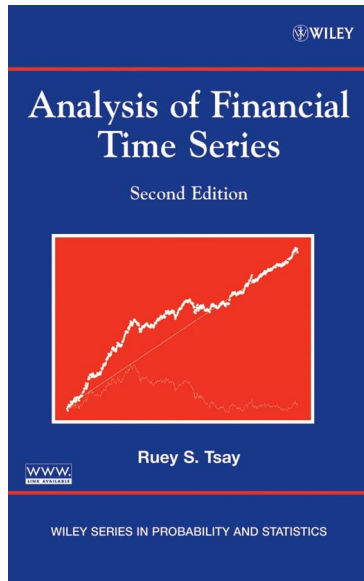
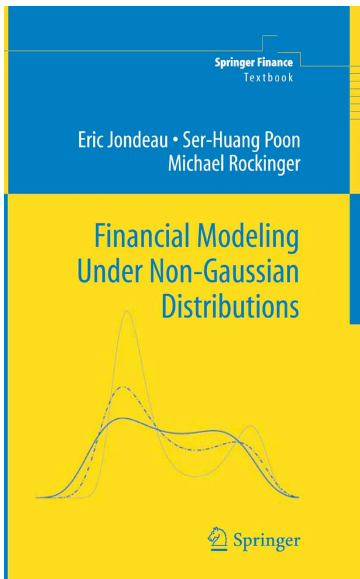
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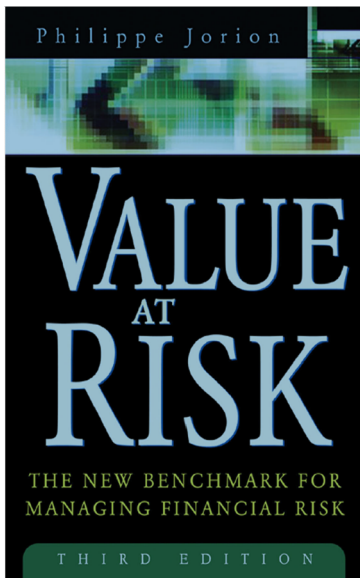
■ Backtesting ES

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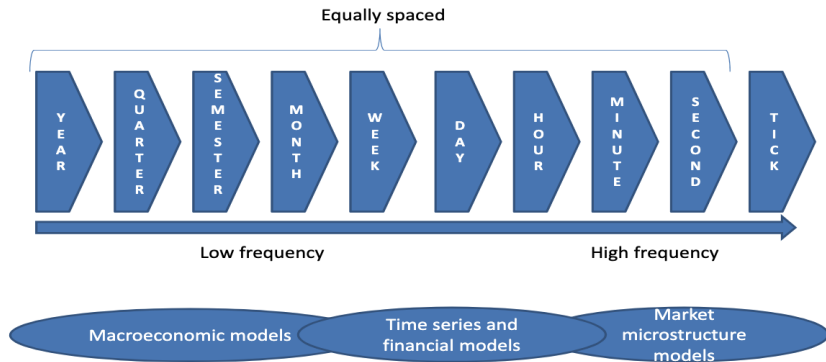
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What is high frequency data? (H.F.D.)

- Twenty years ago, high frequency data referred to **daily financial data**
 - Large data sets consisted of thousands of stocks over 20–30 years (e.g., Center for Research in Security Prices (CRSP) data)
 - Roughly 5–10 million observations, about 100MB of compressed data
- Now, high frequency data refers to **intra-day, tick-by-tick or transaction-level data** on prices, quotes, volume, and the order book
 - Large data sets consist of thousands of stocks over 10–15 years (e.g., New York Stock Exchange (NYSE) TAQ data)
 - Trillions or even more observations, hundreds of terabytes of compressed data

What is high frequency data? (H.F.D.)

Sampling scheme



What is high frequency data? (H.F.D.)

-  **Trading forms:** Equities and ETFs are traded via i) floor trading, ii) limit order book trading, or iii) hybrid systems
- **i) Floor trading data:** Includes information on trades and quotes
 - **ii) Electronic trading data:** Captures order arrivals through the order book
 - ⌚ **Timestamps:** recording time of trades, quotes or when limit orders occur
 - 📊 **Terminology:** Called transactions data, (ultra-)high frequency data, or tick data

High Frequency Data: Basics

- **Sources of high frequency data:** Equity, FX, over-the-counter spot interest rate and forward FX, interest rate futures, bond futures, commodity futures
- **Which markets?** Spot markets, futures, options
- **Where to get high frequency data?** Reuters, Bloomberg, Bridge (over-the-counter and exchanges), exchanges (e.g., LIFFE, TAQ from NYSE), Olsen & Associates (O&A)

Equity - NYSE TAQ

FX - Olsen & Associates

Options - Berkeley Options Database

Fixed Income - GovPX

Suppliers:

Wharton Data Services (WRDS)

Bloomberg, Reuters

OneMarketData OneTick Database (www.onetick.com): equities, fixed income, futures, options and FX.

TICKDATA (www.tickdata.com): equities, fixed income, futures, options and FX.

Thompson-Reuters FastTick

High Frequency Data: Why So Common?

- Data is collected at a very rapid rate — every time new information arrives
 - The original form of financial market data is **tick-by-tick**
 - Practitioners develop trading strategies by observing **high frequency (tick-by-tick)** data
- ⇒ **High frequency data has become the apparatus for understanding market microstructure and financial markets**
- **Challenge:** deal with enormous amount of data and chose the right mathematical tools and models

Challenges to Statistical Modeling

- Huge number of observations
- Dirty data
- Irregularly spaced observations
- Multiple observations with same time stamp
- Heavy-tailed return distributions
- Persistent (Long memory) behavior
- Strong intra-day and intra-week periodicities
- Variables move in discrete increments
- Data for multiple assets seldom occur at the same time

Academic Research Topics

- Market microstructure theory
 - Price discovery and market quality
 - Modeling and estimating liquidity
 - Strategic behavior of market participants
 - Modeling real-time dynamics of trading process
- Event studies
- Estimation and testing of continuous-time models
- **Volatility modeling, estimation and forecasting**

Finance Industry Applications

- Short-term trading
 - Pairs trading
 - Arbitrage strategies
 - Event analysis
- Transaction cost and price impact modeling
 - Order execution
 - Market making
- Derivatives pricing
 - Continuous-time models
 - Volatility estimation
- Risk management

* try to look a bit into the elements in bold on the previous slides

*NEXT SLIDE What This Shows:

Blue bars: Bid sizes at various price levels

Red bars: Ask sizes

Green arrows: Market buy order hitting the ask side

Red arrows: Market sell order hitting the bid side

Mid-price labeled for reference

*An order book is a real-time list of buy and sell limit orders for a specific asset (e.g., a stock). It shows:

Bid side (left): Buyers willing to purchase at certain prices

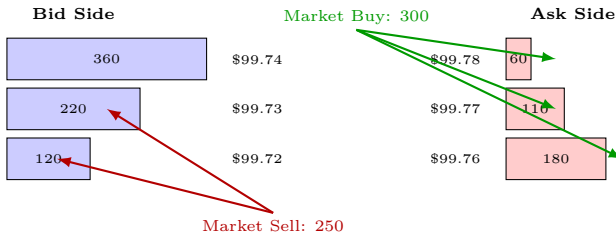
Ask side (right): Sellers offering to sell at certain prices

Each level shows a price and a quantity

High Frequency Data: Example 1

Order Book with Execution Paths

Mid-price: \$99.75



*In the Diagram

Each blue bar is a limit buy order (bids) — higher bar = more volume

Each red bar is a limit sell order (asks)

The mid-price (here, \$99.75) is halfway between the best bid and best ask

Arrows show execution paths of market orders (which consume liquidity)

-Market Buy Order (Green Arrows)

Suppose a trader submits a market buy order for 300 shares.

Market orders execute immediately, so this buy order will:

Hit the lowest ask price first:

\$99.76 for 180 shares

Then move to the next level:

\$99.77 for 110 shares

It needs only 10 more shares, so it partially consumes:

\$99.78 for 10 shares

This is why the green arrows fan out toward the right (ask side).

-Market Sell Order (Red Arrows)

Suppose another trader submits a market sell order for 250 shares.

The sell order hits the best available bids:

\$99.74 for 120 shares

\$99.73 for 130 shares

Total filled: 250 shares

The red arrows show this sell order consuming liquidity on the bid side.

*Summary Concept - Bid Side -Ask Side You want to buy -You hit the asks -
(consume red bars) You want to sell - You hit the bids - (consume blue bars)

Market orders -Execute immediately - At best available price Limit orders -Wait in
the order book -Until matched

High Frequency Data: Example 2

TAQ Database: Trades

SYMBOL	DATE	TIME	PRICE	SIZE	G127	CORR	COND	EX
A	20130305	7:25:39	42.39	200	0	0	T	P
A	20130305	9:11:28	42.03	1700	0	0		D
A	20130305	9:30:00	42.35	47092	0	0	O	N
A	20130305	9:30:00	42.35	100	0	0	F	N
A	20130305	9:30:00	42.32	100	0	0		P
A	20130305	9:30:00	42.34	100	0	0	F	N
A	20130305	9:30:00	42.33	100	0	0	F	P
A	20130305	9:30:00	42.33	100	0	0	F	P
A	20130305	9:30:00	42.34	200	0	0	F	N
A	20130305	9:30:00	42.4	10	0	0	Q	T
A	20130305	9:30:00	42.4	100	0	0	F	T
A	20130305	9:30:00	42.4	100	0	0	F	T
A	20130305	9:30:00	42.42	150	0	0	F	N
A	20130305	9:30:00	42.47	100	0	0	@	N
A	20130305	9:30:00	42.4	100	0	0	F	T
A	20130305	9:30:01	42.42	100	0	0	@	N
A	20130305	9:30:01	42.41	100	0	0	@	N
A	20130305	9:30:02	42.4	100	0	0	@	D
A	20130305	9:30:05	42.46	100	0	0	F	T
A	20130305	9:30:07	42.45	120	0	0	@	D
A	20130305	9:30:07	42.46	100	0	0	F	T
A	20130305	9:30:07	42.46	100	0	0	F	T
A	20130305	9:30:07	42.46	100	0	0	F	N
A	20130305	9:30:07	42.46	100	0	0	@	D

(A: Agilent Technologies Inc)

TAQ Trade Data Field Definitions

- **Time** The time at which the trade entered the Consolidated Trade System (CTS)
Format: cumulative number of seconds since midnight
- **Price** Actual trade price per share. Truncated to four implied decimal places
- **Size** Volume, Number of shares traded
- **Ex** Exchange on which the trade occurred
- **Cond (Sale Condition)** These conditions apply to trades on all exchanges except as indicated
More than one code can be displayed in the field (up to two codes)
- **Corr (Correction Indicator)** Beginning in June 1995, corrections to corrections are permissible
TAQ shows only the original trade and final correction
- **g127** Combined "G", Rule 127, and stopped stock trade indicator
This field applies only to NYSE trades

High Frequency Data: Example 2

TAQ Database: Quotes

SYMBOL	DATE	TIME	BID	OFB	BIDSIZ	OFBSIZ	MODE	EX
A	20130305	4:00:00	0	42.39	0	2	12	P
A	20130305	6:59:45	42.1	42.39	6	2	12	P
A	20130305	7:02:08	42.1	42.38	6	2	12	P
A	20130305	7:04:10	42.1	42.39	6	2	12	P
A	20130305	7:04:45	0	42.39	0	2	12	P
A	20130305	7:25:39	0	45.71	0	22	12	P
A	20130305	8:00:00	0	51.25	0	3	12	T
A	20130305	8:00:00	33.2	51.25	1	3	12	T
A	20130305	8:00:00	0	0	0	0	12	P
A	20130305	8:00:00	33.2	45.71	1	22	12	T
A	20130305	8:00:00	33.2	44.5	1	1	12	T
A	20130305	8:00:00	33.2	44.2	1	2	12	T
A	20130305	8:00:01	39.5	44.2	1	2	12	T
A	20130305	8:00:06	39.4	0	3	0	12	Y
A	20130305	8:00:06	39.4	43.03	3	3	12	Y
A	20130305	8:00:07	39.39	0	9	0	12	Z
A	20130305	8:00:07	39.39	43.03	9	11	12	Z
A	20130305	8:00:53	0	43.28	0	4	12	K
A	20130305	8:00:53	0	43.28	0	7	12	K
A	20130305	8:00:58	39.4	0	3	0	12	Y
A	20130305	8:00:59	39.39	0	9	0	12	Z
A	20130305	8:01:01	39.39	43.03	9	8	12	Z
A	20130305	8:01:01	39.4	43.03	3	2	12	Y
A	20130305	8:01:21	39.39	43.02	9	2	12	Z

(A: Agilent Technologies Inc)

TAQ Quote Data Field Definitions

- **Time** The time at which the quote entered the Consolidated Quote System (CQS)
Format: cumulative number of seconds since midnight
- **Bid** Bid price. Truncated to four implied decimal places
- **Bidsiz**
Bid size in number of round lots (100-share units)
- **Ofr**
Ask price. Truncated to four implied decimal places
- **Ofrsiz**
Ask size in number of round lots (100-share units)
- **Ex**
Exchange on which the quote occurred
- **Mode**
Quote condition. These conditions apply to quotes on all exchanges except as indicated

Eg. of Sale Conditions (Appendix)

- **Regular Way (blank or *)** — Trade made without any stated conditions
- **Cash-Only Basis (A)** — All trading in that issue for that day is on a "cash only" basis
- **Bunched (B)** — Trade representing an aggregate of two or more regular trades (within 60 seconds, at the same price)
- **Cash Sale (C)** — Transaction which calls for the delivery and payment of securities on the same day the trade took place
- **Next Day Settlement Only (D)** — All trading in that issue for that day is for "next day settlement" only
- **NYSE Direct+ (E)** — The high-speed electronic connection for immediate automatic execution (Auto-X) of limit orders up to 1,099 shares
- **BSE Instant Liquidity Access (E)** — Boston Stock Exchange auto execution product, known as ILA
- **Bunched Sold (G)** — Bunched trade not reported within 90 seconds of execution
- **Rule 127 Trade (J)** — (NYSE only) An NYSE trade reported as having been executed as a block position

Other sale conditions

- **L** – Sold Last: Trade occurs in sequence but reported to the tape later.
- **N** – Next Day: Delivery on the first business day after the trade.
- **O** – Opened Last: Opening trade reported late.
- **P** – Prior Reference Price: NASD-only, off-market reference trade.
- **R** – Seller's Option: Seller specifies delivery (2–60 calendar days).
- **S** – Split Trade: NASD-only, trade executed in two markets.
- **T** – Pre/Post-Market Close: Executed outside regular hours but reported same day.
- **U** – Pre/Post-Market Out of Order: Trade timestamped out of market order.
- **W** – Average Price: Price reflects average of multiple trades.
- **Z** – Sold Sale: Reported late with trades occurring between execution and reporting.
- **8/9** – Crossing Session: One-sided (9) or two-sided (8) executions at close.

Corr: Correction Indicator (Appendix)

- **0** — Regular trade that was not corrected, changed, or signified as cancel or error
- **1** — Original trade which was later corrected. This record contains the original time and the corrected data for the trade
- **2** — Symbol correction (out of time sequence)
- **7** — Trade cancelled due to error
- **8** — Trade cancelled
- **9** — Trade cancelled due to symbol correction
- **10** — Cancel record (associated with 8)
- **11** — Error record (associated with 7)
- **12** — Correction record (associated with 1; contains corrected time and original data)

Trade Attributes – G127 and Identifiers (Appendix)

- **"G" Trade** — A member firm trading for its own account must publicly identify that the order is principal
- **Rule 127** — An NYSE trade reported as having been executed as a block position
- **Stopped Stock** — A guarantee by the specialist that an order will be executed at a specific price
- **0** — Does not qualify as "G", Rule 127, or stopped stock trade
- **2** — "G" trade buyer and seller
- **3** — "G" trade buyer
- **4** — "G" trade seller
- **10** — Elect box stroked
- **20** — Percent box stroked
- **40** — A Display Book-reported trade
- **100** — Rule 127 trade
- **200** — Stopped stock trade

TAQ Data Cleaning Rules (Barndorff-Nielsen et al. 2008)

- P1. Delete if time stamp outside the 9:30 am to 4 pm window
- P2. Delete entries with a bid, ask or transaction price equal to zero
- P3. Retain entries originating from a single exchange, delete the others

Quote Data Only

- Q1. When multiple quotes have the same timestamp, replace with a single entry using median bid and ask price
- Q2. Delete entries for which the spread is negative
- Q3. Delete entries for which the spread is > 50 times the median spread on that day
- Q4. Delete entries for which the mid-quote deviates by more than 10 mean absolute deviations from a rolling centered median of 50 observations (excluding the observation itself)

Trade Data Only

- T1. Delete entries with corrected trades (Correction Indicator $\neq 0$)
- T2. Delete entries with abnormal Sale Condition
- T3. If the same time stamp, use the median price
- T4. Delete entries with prices above the ask plus spread, or below the bid minus spread

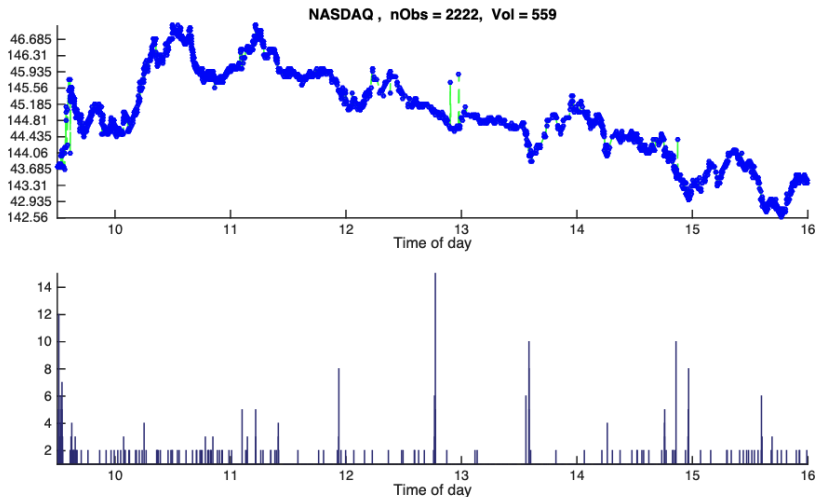
By far the most important rules here are P3, T3 and Q1

We use Q4 to get the outliers that are missed by Q3.

T4 is an attractive rule, as it disciplines the trade data using quotes. However, it has the disadvantage that it cannot be applied when quote data is not available.

In situations where quote data is not available, Q4 can be applied to the transaction prices in place of T4.

Data Cleaning Example : NASDAQ on Jan. 5th, 2000



If multiple transactions have the same time stamp: use the median price

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- Backtesting ES

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Volatility Forecasting Using High-Frequency Data

Two Main Approaches:

■ Reduced-Form Forecasting:

- Time series models (e.g., ARIMA, VAR) on **realized measures**
- Key example - Andersen et al. (2003): long-memory Gaussian VAR
- More recent approaches - Regression-based methods:
 - Heterogeneous Autoregressive - HAR (Corsi, 2009),
 - Mixed-Data Frequency - MIDAS (Ghysels et al., 2006)

■ Model-Based Forecasting:

- Based on return models (e.g., GARCH)
- ⇒ Volatility forecasts inferred from parametric estimation
- High-frequency data used to:
 - Enhance models by relying on realized measures

GARCH is Slow: A Thought Experiment

- Suppose a sudden change in the true latent volatility

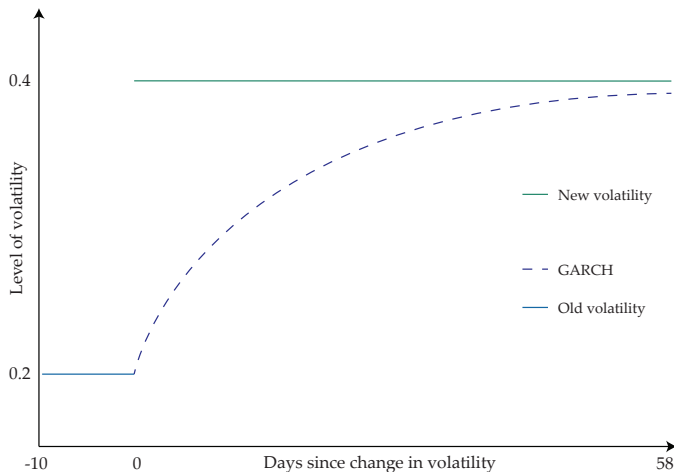
$$\sqrt{\text{var}(r_t|\mathcal{F}_{t-1})} = \begin{cases} 0.2 & t \leq T \\ 0.4 & t > T \end{cases}$$

- With $\alpha + \beta = 1$ (e.g. $\alpha = 0.05$ and $\beta = 0.95$), GARCH(1,1) implies

$$\mathbb{E}(\sigma_{T+k}^2) = \alpha \sum_{j=0}^{\infty} \beta^j \mathbb{E}(r_{T+k-j}^2) = \alpha \sum_{j=0}^{k-1} \beta^j (0.4)^2 + \alpha \sum_{j=k}^{\infty} \beta^j (0.2)^2$$

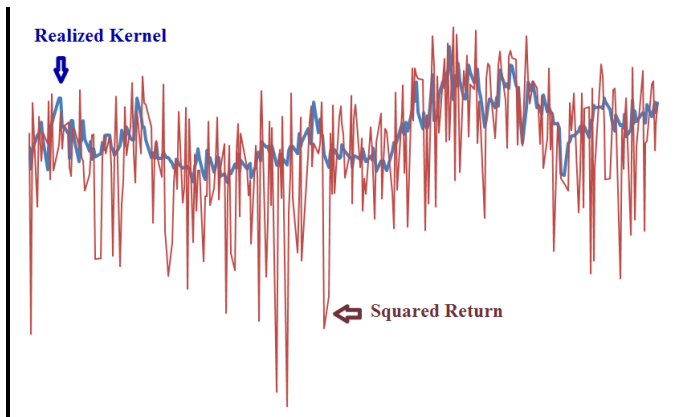
- How quickly does σ_t^2 converge to the new level of $\text{var}(r_t|\mathcal{F}_{t-1})$?

GARCH is Slow



Squared Return is a Noisy Signal of Volatility

- r_t^2 is a very noisy measure of variance
- High-frequency data $\rightarrow$ Realized Measures $\rightarrow$ far more accurate signals





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Realized Measures of Variance

1. Realized Variance
2. Realized Semi-Variance
3. Realized Multipower Variation
4. Realized Kernel

1. Realized Variance

- In practice intraday volatility is unobservable and one only observes asset prices
- Moreover, log price data are only available in discrete time at frequency

$$\Delta\tau = t_i - t_{i-1}$$

where sampling rates $\Delta\tau$ can be seconds or minutes

⇒ For a given sampling rate, $r_{t_i} = p_{t_i} - p_{t_{i-1}}$ are intradaily log-returns

- We assume M intradaily observations $t - 1 = t_0 < t_1 < \dots < t_M = t$

⇒ the daily **Realized Variance (RV)** is given by

$$RV_t = \sum_{i=1}^M r_{t_i}^2$$

and one can show that **in absence of jumps** it is a good estimator of the variance

Link with continuous-time price models

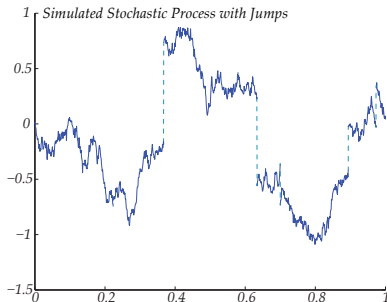
- Financial observations are often sampled at high frequency
- ⇒ to the limit, infinitesimal differences can be considered :

$$dt = (t + h) - t, \quad h \rightarrow 0$$

- ⇒ Applied to prices S_t , infinitesimal calculus comes down to continuous-time stochastic differential equation dS_t

But Abrupt changes can occur in price dynamics

- ⇒ such jumps in price cannot be modeled by Gaussian increments



Fundamental theorem of asset pricing:

Under the EMH the market exists but risk free profit is impossible $\Rightarrow$ if there exists a risk-neutral probability measure $\mathbb{Q}$ equivalent to the original probability measure $\mathbb{P}$ under which S is a σ -martingale, then S does not permit free lunch with vanishing risk

$\Rightarrow$ option pricing; use Itô's lemma to solve the stochastic differential equation when assuming a Itô drift-diffusion for the returns of S_t

E.g. - Ornstein-Uhlenbeck (OU) model = continuous time counterpart of an AR(1) process

- Hull-White (HW) model
- Cox-Ingersoll-Ross (CIR) model (mimics the evolution of interest rates)
- Heston 1993 : two-factor SV that allows for time-varying spot volatility σ
- Bates (1996) extends the Heston model to Poisson jump-diffusion

All in all: Continuous-time models are central to mathematical finance and Most theoretical results on derivative pricing rely on continuous-time processes, obtained as solutions of diffusion equations

Aim: Construct continuous time models with features of GARCH (it is possible to establish connections between the two approaches: discrete-time and continuous-time models)

Note: In continuous time it is natural to model the logarithm of the asset price itself, that is p_t , rather than its increments r_t as in discrete time

Link with continuous-time price models

Definition

Over an interval of time $[t-1, t]$ (generally fixed to one day) the log-return variance is called **Integrated Variance, IV** when prices follow a **continuous process**

Definition

Over an interval of time $[t-1, t]$ (generally fixed to one day) the log-return variance is called **Quadratic Variation, QV** when prices follow a **continuous process contaminated by jumps**

Counting processes $\mathcal{CM}$ are good candidates

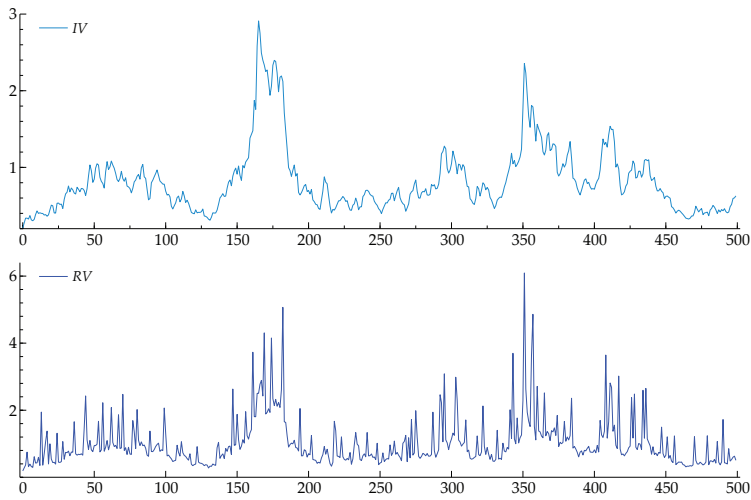
A counting process is a stochastic non-decreasing process $N_t \in \mathbb{N}$, $t \geq 0$ [$\Rightarrow$] As $N_t \in \mathbb{N}$, for $s \leq t$, $N_t - N_s$ is the number of events occurred in $(s, t]$

NEXT SLIDE:

RV is noisy, especially in presence of umps as it is not appropriate for estimating
IV

See later BV, a robust measure

1. Realized Variance



2. Realized Semi-Variance

- The behavior of r_t in the lower tail of their distribution is of interest
... as it often differs from the behavior in the upper tail
⇒ requires an asymmetric treatment of the downside and upside risks
- Barndorff-Nielsen et al. (2010) suggest the [Realized Semi-variance](#)

$$RS_t^+ = \sum_{i=1}^M r_{t_i}^2 \mathbb{1}_{r_{t_i} \geq 0} \quad \text{and} \quad RS_t^- = \sum_{i=1}^M r_{t_i}^2 \mathbb{1}_{r_{t_i} \leq 0}$$

3. Realized Multipower Variation

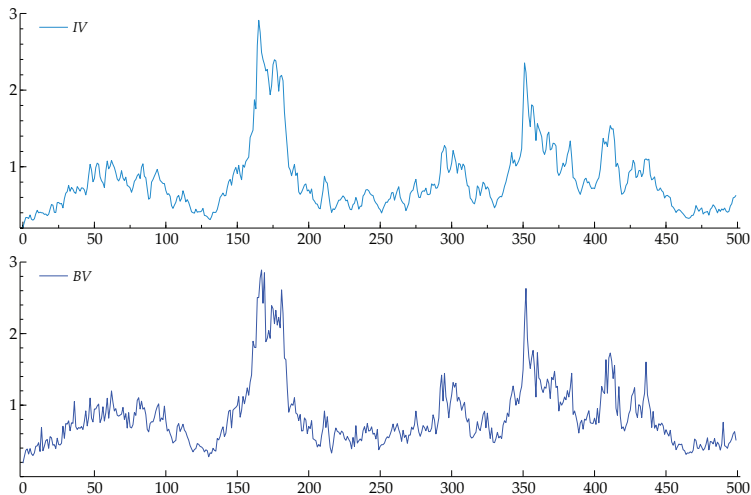
- In presence of jumps, $RV_t \xrightarrow{p} QV_t$ instead of the IV_t as $M \rightarrow \infty$
- As jumps cannot be easily distinguished from the continuous part,
... if IV_t is of interest, **robust realized measures** are needed
- ⇒ Barndorff-Nielsen et al. (2006) introduce the Multipower Variation
- The simplest case is the **Bipower Variation (BV)**

$$BV_t^{\{1,1\}} = BV_t = \frac{\pi}{2} \sum_{j=2}^M |r_{t_j}| |r_{t_{j-1}}|$$

- Conversely to RV_t , this measure is jump robust as $M \rightarrow \infty$:

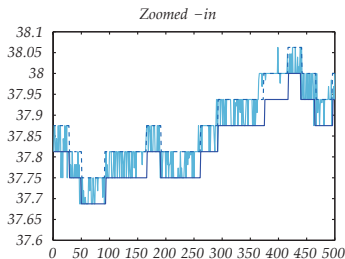
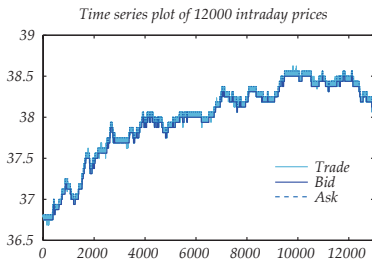
$$BV_t \xrightarrow{p} IV_t$$

3. Bipower Variation



Microstructure noise

- The most striking feature of the empirical prices is their discreteness
- ⇒ price movement cannot be smaller than one tick



- ⇒ efficient log-prices (from theoretical continuous-time models) are unobservable due to [microstructure noise](#), i.e.

- frictions in the trading process : tick limit, rounding error
- informational effects : gradual response of prices to a block trade
- recording errors : prices entered as zero or misplaced decimal points

4. Realized Kernel

- Let $p_t^* = p_t + \varepsilon_t$ be the observed price

$\Rightarrow \varepsilon_t \perp p_t$ acts as microstructure noise and is defined as a white noise

- Barndorff-Nielsen et al. (2008) suggest the **Realized Kernel (RK)**

$$RK_t = \underbrace{\gamma_0(p_t^*)}_{RV} + \underbrace{\sum_{h=1}^H \kappa\left(\frac{h}{H}\right) (\gamma_h(p_t^*) + \gamma_{-h}(p_t^*))}_{\text{noise correction}} \xrightarrow{p} QV_t$$

with

$\gamma_h(p_t^*) = \sum_{j=1}^M (p_{t_j}^* - p_{t_{j-1}}^*)(p_{t_{j-h}}^* - p_{t_{j-1-h}}^*)$ the return autocovariance function and

$\mathcal{K}(\cdot)$ a kernel weighting function

Kernel = weighting function

The recommended kernel $\mathcal{K}(\cdot)$ is Parzen's kernel

$$\mathcal{K}(x) = \begin{cases} 1 - 6x^2 + 6x^3, & 0 > x \geq \frac{1}{2} \\ 2(1 - x)^3, & \frac{1}{2} > x \geq 1 \\ 0, & x > 1 \end{cases}$$

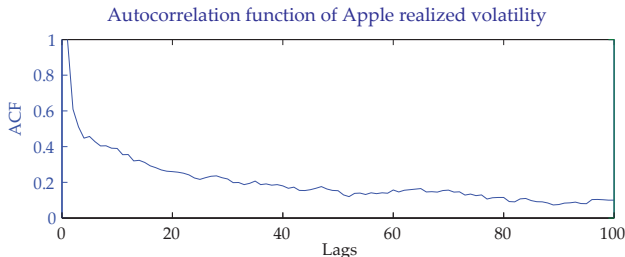
The preferred choice for the bandwidth is $H = 3.5134 \times \hat{\xi}^{4/5} M^{3/5}$ with

$$\hat{\xi} = q^{-1} \sum_{i=1}^q \hat{\omega}_{(i)}^2 / RV_{\text{sparse}} \text{ and } \hat{\omega}_{(i)}^2 = RV_{\text{dense}}^{(i)} / (2M_{(i)})$$

where RV_{sparse} ($RV_{\text{dense}}^{(i)}$) are based on 20 min (2 min independent) r_{t_i}

Realized measures of variance

- RV_t , RS_t , BV_t and RK_t are **ex-post, realized** measures of volatility ($\mathcal{RM}$)
- ... at the opposite, GARCH-type models estimate **ex-ante, conditional** volatility
- Realized measures are **nonparametric estimators** of IV_t or QV_t
- ⇒ without conditional forecasting models, parametric forecasts using realized measures are infeasible



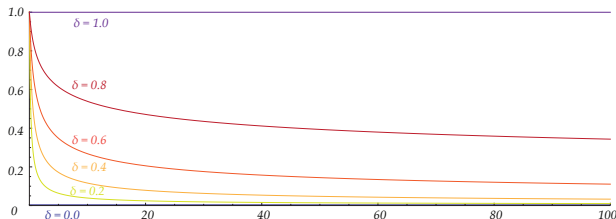
- $\mathcal{RM}$ are highly persistent with hyperbolic decay of autocorrelations
- ⇒ one could use this stylized fact to **forecast volatility**

Modelling realized measures with ARFIMA

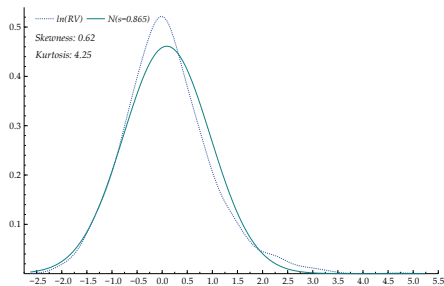
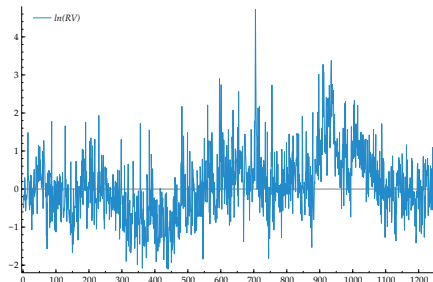
- Autocorrelations of ARMA models decay at exponential rate
 $\Rightarrow$ only $\text{ARMA}(\infty)$ can mimic an hyperbolic decay
- Such a slow decay reflects the presence of **Long Memory** ($\mathcal{LM}$)
 $\Rightarrow$ the simplest parametric model of $\mathcal{LM}$ is the $\text{ARFIMA}(p, \delta, q)$

$$(1-L)^\delta \underbrace{\Phi(L)(x_t - \mu) = \Theta(L)\varepsilon_t}_{\text{ARMA}(p,q)} \Rightarrow \varepsilon_t = \sum_{i=0}^{\infty} \Xi_i x_{t-i}$$

with $\delta \in (-1/2, 1)$ and $x_t \in \log(\mathcal{RM})$ or $x_t \in (\mathcal{RM})^{1/2}$



Realized Variance: normality hypothesis



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i. Forecasting realized measures with HAR

- Corsi (2008) suggests a **Heterogeneous AR** model that mimics $\mathcal{LM}$

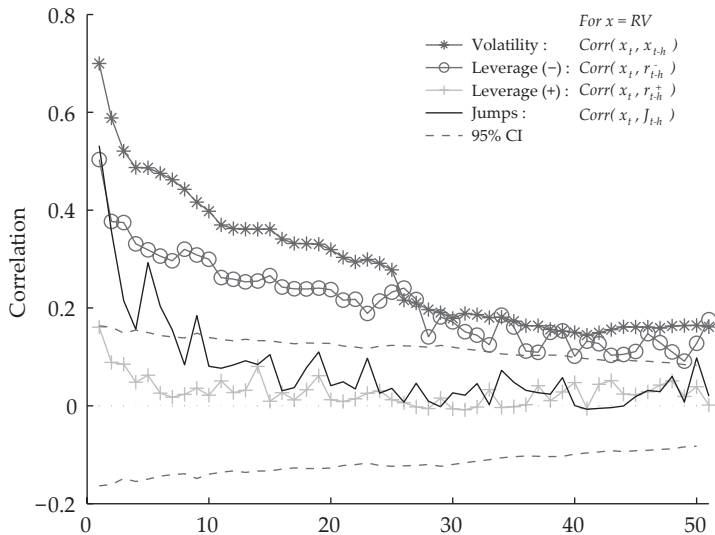
$$x_{t+1} = c + \underbrace{\beta_d x_t}_{\text{day}} + \underbrace{\beta_w x_t^w}_{\text{week}} + \underbrace{\beta_m x_t^m}_{\text{month}} + \varepsilon_{t+1}, \quad \varepsilon_t \sim \text{i.i.d.},$$

where $x_t^w = \frac{1}{5} \sum_{j=1}^4 x_{t-j}$ and $x_t^m = \frac{1}{22} \sum_{j=1}^{21} x_{t-j}$ and $x_t \in (\mathcal{RM})^{1/2}$

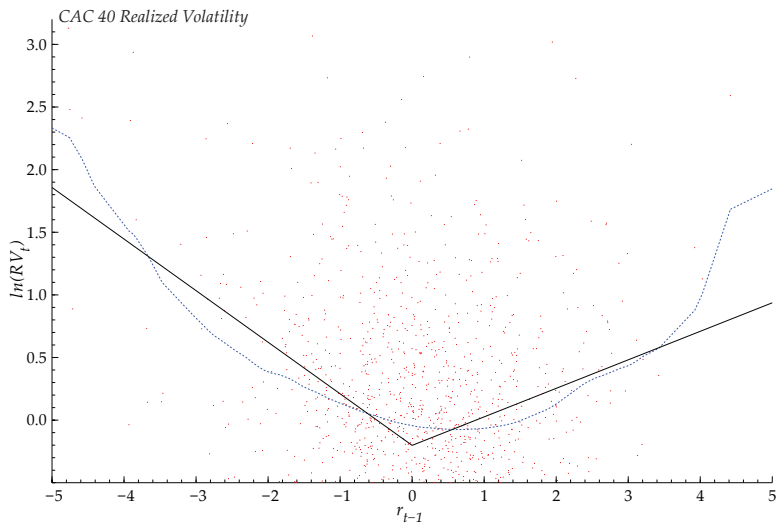
- A log-HAR version is possible if $x_t \in \log(\mathcal{RM})$

Leverage in Realized Variance

The HAR is simple and flexible $\Rightarrow$ can model additional stylized facts



Leverage in Realized Variance



Extensions of the HAR model : LHAR (Appendix)

- Define aggregated negative and positive returns at given frequency s

$$(r_t^s)^- = \frac{1}{s} \sum_{j=1}^s r_{t-j} \mathbb{1}_{\sum_{j=1}^s r_{t-j} < 0} \text{ and } (r_t^s)^+ = \frac{1}{s} \sum_{j=1}^s r_{t-j} \mathbb{1}_{\sum_{j=1}^s r_{t-j} \geq 0}$$

- We allow for the leverage effect to impact each market component

$$\begin{aligned} x_{t+1} = & c + \beta_d x_t + \beta_w x_t^w + \beta_m x_t^m \\ & + \gamma_d^- (r_t)^- + \gamma_w^- (r_t^w)^- + \gamma_m^- (r_t^m)^- \\ & + \gamma_d^+ (r_t)^+ + \gamma_w^+ (r_t^w)^+ + \gamma_m^+ (r_t^m)^+ + \varepsilon_{t+1} \end{aligned}$$

⇒ The Leverage HAR (LHAR) fits particularly well the asymmetric relationship between r_t and x_{t+1} with $x_t \in (\mathcal{RM})^{1/2}$

Extensions of the HAR model : LHAR-CJ (Appendix)

- Assume now that J_{t,T_M} is the sequence of significant realized jumps
- In the spirit of the HAR let's define at given frequency s

$$J_{t,T_M}^s = \frac{1}{s} \sum_{j=1}^s J_{t-j,T_M}$$

- The Leverage HAR with Continuous volatility and Jumps (LHAR-CJ) is given by

$$\begin{aligned} x_{t+1} = & c + \beta_d x_t + \beta_w x_t^w + \beta_m x_t^m \\ & + \alpha^d J_{t,T_M} + \alpha^w J_{t,T_M}^w + \alpha^m J_{t,T_M}^m \\ & + \gamma_d^- (r_t)^- + \gamma_w^- (r_t^w)^- + \gamma_m^- (r_t^m)^- \\ & + \gamma_d^+ (r_t)^+ + \gamma_w^+ (r_t^w)^+ + \gamma_m^+ (r_t^m)^+ + \varepsilon_{t+1} \end{aligned}$$

- As the model is linear, one can use OLS to estimate all parameters
- This general model offers good in-sample and out-of-sample performances

ii. GARCH-X

- Engle (2002) (and many others) used GARCH-X models

$$\sigma_t^2 = \omega + \alpha r_{t-1}^2 + \beta \sigma_{t-1}^2 + \gamma x_{t-1}$$

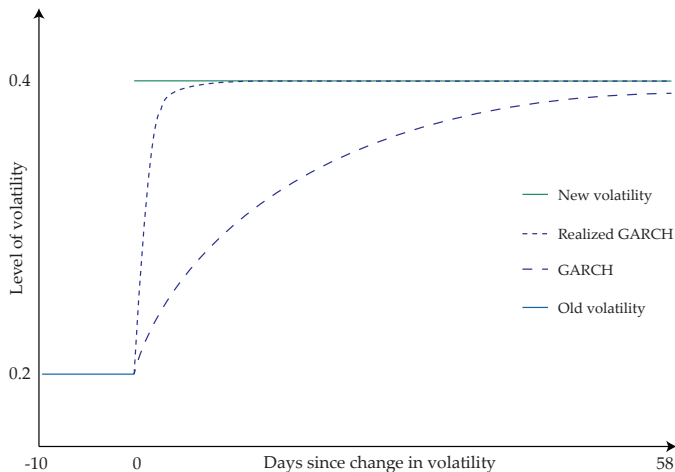
$x_t \in \mathcal{RM}$ is a realized measure of the QV (not IV)

- Leads to substantial empirical gains

⇒ Typically

- $\hat{\gamma} \simeq 0.5$
- $\hat{\alpha} \simeq 0$ (ARCH parameter becomes insignificant)
- However, GARCH-X models are partial (incomplete) models that have nothing to say about returns and volatility beyond a single period into the future

iii. GARCH with a Realized Measure is Fast



iii. Realized GARCH (Hansen, Huang, Shek, 2012)

- The Realized GARCH is a complete model, contrary to GARCH-X

⇒ It fully specifies the dynamic properties of both r_t and x_t :

$$\begin{aligned}
 r_t &= \overbrace{\mu + \sqrt{\sigma_t^2} z_t}^{\text{Return Equation}} \\
 \log \sigma_t^2 &= \alpha + \beta \log \sigma_{t-1}^2 + \gamma \log x_{t-1} \} \text{ GARCH Equation} \\
 \log x_t &= \underbrace{\xi + \psi \log \sigma_t^2 + \tau(z_t) + u_t}_{\text{Measurement Equation}}
 \end{aligned}$$

- $x_t \in \mathcal{RM}$ is a realized measure of the QV (not IV)
- $\tau(z)$ models an asymmetric response in volatility to return shocks (leverage effect)
- $z_t \sim \text{i.i.d. } (0, 1), u_t \sim \text{i.i.d. } (0, \sigma_u^2)$
- $\tau(z_t) + u_t$ are the volatility shocks

iii. Realized GARCH multi-period forecast

- Predicts both the conditional return variance and the realized measure

e.g. RGARCH(1,1) forecasts through the VARMA(1,1) structure

$$\begin{pmatrix} \log \sigma_t^2 \\ \log x_t \end{pmatrix} = \begin{pmatrix} \beta & \gamma \\ \psi\beta & \psi\gamma \end{pmatrix} \begin{pmatrix} \log \sigma_{t-1}^2 \\ \log x_{t-1} \end{pmatrix} + \begin{pmatrix} \alpha \\ \psi\alpha + \xi \end{pmatrix} + \begin{pmatrix} 0 \\ \tau(z_t) + u_t \end{pmatrix}$$

- h -step ahead forecasts are obtained from

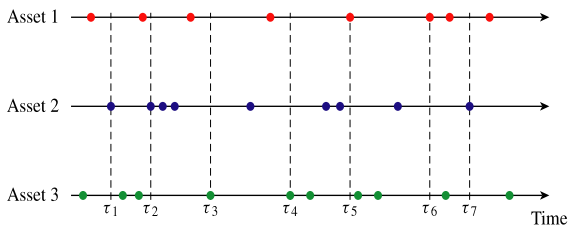
$$\begin{pmatrix} \widehat{\log \sigma_{t+h}^2} \\ \widehat{\log x_{t+h}} \end{pmatrix} = \begin{pmatrix} \beta & \gamma \\ \psi\beta & \psi\gamma \end{pmatrix}^h \begin{pmatrix} \log \sigma_{t-1}^2 \\ \log x_{t-1} \end{pmatrix} + \sum_{j=0}^{h-1} \begin{pmatrix} \beta & \gamma \\ \psi\beta & \psi\gamma \end{pmatrix}^j \left(\begin{pmatrix} \alpha \\ \psi\alpha + \xi \end{pmatrix} + \begin{pmatrix} 0 \\ \tau(z_{t+h-j}) + u_{t+h-j} \end{pmatrix} \right)$$

- One would have to account for distributional aspects of $\log \sigma_{t+h}^2$ in order to produce an unbiased forecast of it since $\mathbb{E}(\log x) \neq \log(\mathbb{E}(x))$

iv. Extensions: Multivariate realized measures

1. Realized Covariance (Barndorff-Nielsen and Shephard, 2004)
2. Realized Semi-Covariance (Bollerslev, Patton, Quaedvlieg, 2020)
3. Realized Bipower Covariation (Barndorff-Nielsen and Shephard, 2004b)
4. Realized Kernel (Barndorff-Nielsen, Hansen, Lunde, and Shephard, 2011)

Asynchronous trading issue: The asynchronous nature of intraday prices biases realized covariances toward 0, unless an appropriate adjustment is made



previous tick aggregation : forces prices to an equispaced grid by taking the last price realized before each grid point or, alternatively, the interpolation of the 1st and last price in the interval

But, at least one quote should be available for both assets in the chosen time interval for this algorithm to be applicable

Empirical work use heuristically chosen 5 or 30 minutes return interval to try to avoid the bias and market microstructure effects

A better approach: **Refresh-time sampling (Harris and Wood, 1995)**

forces the time series to synchronize but not necessarily on equispaced time grid

iv. Extensions: Multivariate realized measures

Realized correlation and realized beta

- Realized Correlation is the realized analogue of the usual correlation estimator, but defined in terms of realized covariance (RC)

$$RCorr = \frac{RC_{vl}}{\sqrt{RC_{vv}RC_{ll}}}$$

- Suppose

$$RC = \begin{pmatrix} RV_v & RC'_{fv} \\ RC_{fv} & RC_{ff} \end{pmatrix}$$

is the realized covariance matrix of an asset with a set of observable factors. The realized beta is defined as

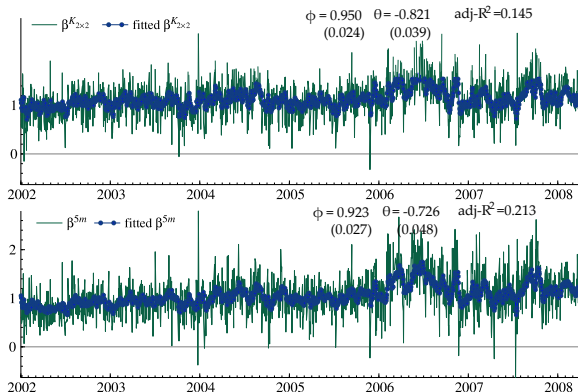
$$R\beta = RC_{ff}^{-1}RC_{fv}$$

- Realized betas are similar to other realized measures in that they are model free and, as long as prices can be sampled frequently and have little market microstructure noise, is an accurate measure of the current exposure to changes in the market

Illustration: Realized correlation and realized beta

Stylized facts (AA <Alcoa Inc.> - SPY analysis)

- Estimate the realized kernel beta and 5 min-based realized betas
- Model them as ARMA(1,1) processes



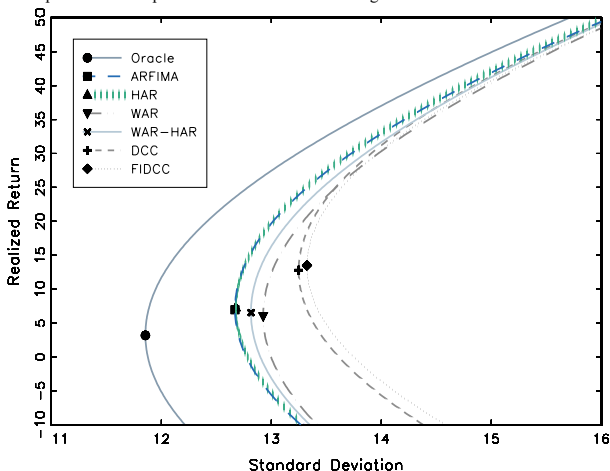
Barndorff-Nielsen, O. E., Hansen, P. R., Lunde, A., & Shephard, N. (2011). Multivariate realised kernels: consistent positive semi-definite estimators of the covariation of equity prices with noise and non-synchronous trading. *Journal of Econometrics*, 162(2), 149-169.

Both models have a significant memory, with autoregressive roots well above 0.9 and with large moving average roots. The fit of the realised kernel beta is a little bit better than that for the realised beta.

Barndorff Nielsen et al JOE 2011.

Illustration: Economic value of volatility forecasts : portfolio optimization

Mean-variance plots for the ex-post realized conditional mean against realized conditional standard deviation



All plots are averages across the 648 out-of-sample periods (days)

Chiriac, R., & Voev, V. (2011). Modelling and forecasting multivariate realized volatility. *Journal of Applied Econometrics*, 26(6), 922-947.

CHIRIAC AND VOEV 2011 PAPER , SEE HFE SLIDES ORLEANS

oracle forecast: the cumulated RCov matrix computed over the actual period $t : t + s$
oracle forecast has a superior mean–variance trade-off

VARFIMA and HAR-based plots almost overlap (similar performance)

A natural point to be considered is the level of return corresponding to the global minimum variance portfolio (GMVP), accentuated in Figure 1 by a particular symbol for each model.

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Market risk

Definition

Market risk is the risk that the value of a trading portfolio of a bank will decrease (loss of a bank) due to the change in value of the risk factors (FRTB October 2013)

- Distinguish: Equity risk, Foreign exchange risk, Commodity risk, Interest-rate risk, Credit spread risk
- The goal of risk measurements is to provide investors with a means to aid investment decision making
- Since the 1950s when financial theory emerged, investors have desired a number or quantitative measure which can be used to approximate an asset's risk
- Since Basel II, Value-at-Risk is the most used statistical measure of downside risk
- Since Basel III, Expected Shortfall has become compulsory

- Markovitzian financial theory suggests that investors consider an asset's ability to deviate from expected returns as an indication of the risk of holding such an asset
- risk of losses (drop in the value of a trading portfolio) owing to movements in the level or volatility of risk factors
- Absolute vs. relative risk (in dollars - total returns - vs. relative to a benchmark index - as a tracking error, deviation)
- dir - exposure to a direction of movement in a financial variable (stock, exchange, commodity, interest rate) vs. non-linear exposures and exposures to hedged positions or to volatilities

-The Basel Committee gives a very precise list of risk factors by asset classes (FRM lecture 2)

For instance, the equity delta risk factors are the equity spot prices and the equity repo rates, the equity vega risk factors are the implied volatilities of options, and the equity curvature risk factors are the equity spot prices

In the case of the interest rate risk class (GIRR), the risk factors include the yield curve, at curve of market-implied inflation rates for each currency and some cross-currency basis risks.

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Broad measures of market risk

- 1 **Volatility:** a measure of the magnitude of variations of an asset value (deviations from the mean) – volatility is the second moment of probability distribution (**standard deviation**)
- 2 **Value-at-Risk (VaR)**
- 3 **Expected Shortfall (ES), or Conditional VaR**

Losses can occur through a combination of two factors: the volatility in the underlying financial variable and the exposure to this source of risk. Whereas corporations have no control over the volatility of financial variables, they can adjust their exposure to these risks, for instance, through derivatives. Value at risk (VAR) captures the combined effect of underlying volatility and exposure to financial risks.

Volat and correlation are not fungible, and it is not possible to convert them into potential losses

value of potential losses in monetary value which synthesize all risk metrics in a single potential loss figure

Because VaR is synthetic it is not a replacement for such specific measures, but it summarizes them and provides a global view.

Drawback:

Without providing an insight on the sources of risk, it does tell which factors a portfolio is mostly sensitive to.

Because it synthesizes risk, it is not adequate for risk management, except for setting up limits on aggregated VaR. But since VaR models do not guarantee backward traceability to sources of risks, it is not suited for other than global limit risk management.

1. Volatility

- Measures the magnitude of deviations from the mean of a market variable, i.e. standard deviation
- Volatility and correlation forecasts are necessary for valuing derivatives
- Is unobservable and generally calculated for (logarithmic) returns as **equally weighted sample historical volatility**, but they are quite unstable (their variations exceed sampling error)

$$\sigma^2 = \frac{1}{T-1} \sum_{i=1}^T (r_i - \bar{r})^2$$

with r_i the logarithmic returns

- **Time-varying volatilities** are much better:
 - i) GARCH based parametric measures
 - ii) Realized measures
- Distinguish also **implicit volatilities**, embedded in option prices

- Sensitivities only measure the response of a market variable to a given shift of risk factors, but they ignore the volatility of risk factors
- Combining sensitivity with volatility provides a more comprehensive view of how unstable a market variable is depending on risk factors.
- the “true” value of volatility is not observable. The sampling error decreases with the number of observations
- **historical volatility** assumes the random variable follows a stochastic process through time that this is identically independently distributed, or “i.i.d.,” with, notably, constant variance.
- σ^2 unbiased estimate of variance; but if $t - 1$ is replaced by t it becomes the ML estimator
- TV volatility : in crisis periods there are volatility spikes, so we account for the stochastic behaviour of volatility by using for eg GARCH models instead of constant variance assumptions
- **Implicit volatilities** are embedded in the price of options. It is obtained by inverting the Black-Scholes formula, looking for which value of volatility matches observed prices. Implied volatilities in option prices vary depending on strike and maturity (“the volatility surface”) at any given date. Series of implied volatility are calculated with at-the-money options. Implicit volatilities look forward by definition, as market prices do, but they might be very unstable.

Equally weighted sample historical volatility

- Provides the long-term estimate, “unconditional”, volatility, but it is not representative of current conditions
- Regulations require comparing volatilities over 60 days and over 250 days and using 10-day volatilities
- Assume logarithmic returns $r_i = \ln(V_{i+1}/V_i)$, $i = 1, 2, \dots, n$
- Under the i.i.d. returns hypothesis, continuous expected return is constant and has constant variance σ^2
- Overall, $\sigma_n^2 = \sigma_1^2 \times n$, or, equivalently, $\sigma_n = \sigma_1 \times \sqrt{n}$
- For example, the 10-day volatility, the horizon used for regulations for market risk, would be obtained by scaling by $\sqrt{10} = 3.162$, the daily volatility
- Issues: independence and stationarity hypotheses

- The variance grows proportionally to time and the volatility grows proportionally to the square root of time. This is summarized by the statement: “uncertainty grows as the square root of time,”...

-So, in practice, one scales up daily volatility over any time horizon, as long as the i.i.d. assumption holds.

-60 and 250 = sample size for computing daily volatility that is afterwards scaled to 10-day

- The historical volatility, under i.i.d. assumptions, can be calculated using moving windows of time periods, see later for historical VaR.

— **e.g. for students** The annualized historical volatility, starting from daily volatilities, is $\sigma(250) = \sigma(1)\sqrt{250} = 15.8114\sigma(1)$,

— **e.g. for students** Assume that the daily volatility or daily standard deviation of stock index return for the last 250 working days is 2%.

The 10-day volatility is $\sigma_{10} = \sigma_1\sqrt{10} = 2\% \times 3.16 = 6.32\%$ and annualized volatility is $\sigma_{250} = 2\% \times 15.81 = 31.62\%$

Reminder: GARCH Models

- Capture the time-varying and mean-reverting behaviour of volatility by combining a long-term variance rate with the last estimate of the variance and the last return
- GARCH(1,1) equation (Bollerslev 1986):

$$\sigma_t^2 = \omega + \alpha r_{t-1}^2 + \beta \sigma_{t-1}^2$$

- The distribution of the return is not identically distributed anymore since its variance now depends on the date t , therefore labeled “conditional variance”
- $\omega = \gamma * \sigma_L^2$, where σ_L^2 stands for the unconditional, long-term variance and $\gamma = 1 - \alpha - \beta$
- The model is estimated by ML under stationarity and positivity constraints (see Time-Series course in M1)

mean reversion

The GARCH (1,1) model recognizes that over time the variance tends to get pulled back to a long-run average level of σ_L^2

The GARCH(1,1) is equivalent to a model where the variance V follows the stochastic process

$$dV = a(V_L - V)dt + \xi V dz,$$

with $a = 1 - \alpha - \beta$, $\xi = \alpha\sqrt{2}$, which is a mean-reverting process.

The variance has a drift that pulls it back to V_L at rate a . When $V > V_L$, the variance has a negative drift; when $V < V_L$, it has a positive drift. Superimposed on the drift is a volatility ξ

In practice, variance rates do tend to be mean reverting. The GARCH(1,1) model incorporates mean reversion, whereas the EWMA model does not. GARCH (1,1) is therefore theoretically more appealing than the EWMA model.

GARCH Models : example

- Suppose a GARCH(1,1) model estimated from daily data as

$$\sigma_t^2 = 0.000002 + 0.13r_{t-1}^2 + 0.86\sigma_{t-1}^2$$

1. Compute the long-run average volatility per day implied by the model

- Identify: $\alpha = 0.130$, $\beta = 0.86$, $\omega = 0.000002$
- Since $\gamma = 1 - \alpha - \beta$, it follows that $\gamma = 0.01$
- From $\omega = \gamma\sigma_L^2$ one obtains $\sigma_L^2 = 0.0002$ and $\sigma_L = 0.014 = 1.4\%$ per day

2. Compute the new estimate of volatility if the estimate of volatility at $t - 1$ is 1.6% per day and the market variable decreased by 1%

$$\Rightarrow \sigma_{t-1}^2 = 0.016^2 \text{ and } r_{t-1}^2 = 0.01^2$$

- Then $\sigma_t^2 = 0.000002 + 0.13 \times 0.01^2 + 0.86 \times 0.016^2 = 0.00023516$
- And $\sigma_t = 0.0153$

Empirical example: mean-reversion

Reminder: GARCH forecasting

- For a GARCH(1,1) model

$$y_t = \mu + \varepsilon_t, \quad \varepsilon_t \sim N(0, \sigma_t^2), \quad \sigma_t^2 = \alpha_0 + \alpha_1 u_{t-1}^2 + \beta \sigma_{t-1}^2$$

- the one-period-ahead forecast is

$$\sigma_{1,T}^{2f} = \alpha_0 + \alpha_1 u_T^2 + \beta \sigma_T^2$$

- the two period-ahead forecast is

$$\sigma_{2,T}^{2f} = \alpha_0 + \alpha_1 \mathbb{E}(u_{T+1}^2 | \mathcal{F}_T) + \beta \sigma_{1,T}^{2f} \quad (1)$$

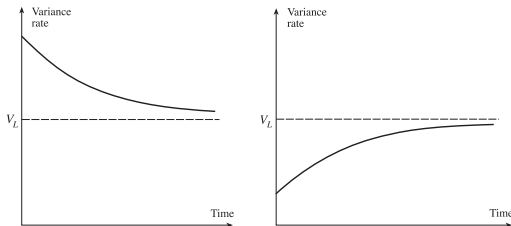
$$= \alpha_0 + (\alpha_1 + \beta) \sigma_{1,T}^{2f} \quad (2)$$

- and, more generally, the $s \geq 2$ periods-ahead forecasts are

$$\sigma_{s,T}^{2f} = \alpha_0 \sum_{i=1}^{s-1} (\alpha_1 + \beta)^{i-1} + (\alpha_1 + \beta)^{s-1} \sigma_{1,T}^{2f}$$

Empirical example: mean-reversion

Figure 23.3 Expected path for the variance rate when (a) current variance rate is above long-term variance rate and (b) current variance rate is below long-term variance rate.



- GARCH(1,1) results: $\alpha + \beta = 0.9935$ and $V_L^{(b)} = 0.0002075$
- Suppose that the estimate of the current variance rate per day is 0.0003.
- In 10 days, the expected variance rate is $0.0002075 + 0.9935^{10}(0.0003 - 0.0002075) = 0.0002942$ (exp. vol. per day is 1.72%)
- The expected variance rate in 500 days is $0.0002075 + 0.9935^{500}(0.0003 - 0.0002075) = 0.0002110$

Figure 23.3 shows the expected path followed by the variance rate for situations where the current variance rate is different from V_L . As mentioned earlier, the variance rate exhibits mean reversion with a reversion level of V_L and a reversion rate of $1 - \alpha - \beta$. Our forecast of the future variance rate tends towards V_L as we look further and further ahead. This analysis emphasizes the point that we must have $\alpha + \beta < 1$ for a stable GARCH(1,1) process.

When $\alpha + \beta > 1$, the weight given to the long-term average variance is negative and the process is “mean fleeing” rather than “mean reverting”.

For the S&P 500 data considered earlier, $\alpha + \beta = 0.9935$ and $V_L = 0.0002075$. Suppose that the estimate of the current variance rate per day is 0.0003. (This corresponds to a volatility of 1.732% per day.) In 10 days, the expected variance rate is $0.0002075 + 0.9935^{10}(0.0003 - 0.0002075) = 0.0002942$

The expected volatility per day is 1.72%, still well above the long-term volatility of 1.44% per day. However, the expected variance rate in 500 days is $0.0002075 + 0.9935^{500}(0.0003 - 0.0002075) = 0.0002110$ and the expected volatility per day is 1.45%, very close to the long-term volatility.

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Value-at-Risk

Definition (Jorion (2007))

“How bad can things get?": VaR summarizes the worst loss over a target horizon that will not be exceeded with a given level of confidence.

Definition (Tsay (2012))

VaR is defined as a single estimate of the amount by which an institution / investor's position in a risk category could decline because of general market movements during a given holding period

- Main tool to measure downside market risk based on current positions
- Main advantage: it summarizes risk in a single, easy-to-understand number
- Black Swans do not appear in past data, and thus will likely not feature in predicted risk estimates such as VaR!

Jorion, Philippe. Financial risk manager handbook. Vol. 406. John Wiley & Sons, 2007.

Tsay, Ruey S. An Introduction to Analysis of Financial Data with R. John Wiley & Sons, 2012.

Value-at-Risk

- VaR is the worse potential loss over a given time horizon such that there is a low, pre-specified probability that the actual loss will be larger
- e.g. A bank might announce that the daily VaR on its trading portfolio is \$50 million at the 99% CI. $\Rightarrow$ There is only one chance in 100, under normal market conditions, for a loss greater than \$50 million to occur.

it is easy to understand. If we accept that it is useful to have a single number to describe the risk of a portfolio, an interesting question is whether VaR is the best alternative. – see the inconvenients at the end of the section

general methodology: defining percentiles of the random target variable (e.g. the P&L for market risk or credit losses for credit risk), which has become a common tool in risk methodology, independently of VaR itself.

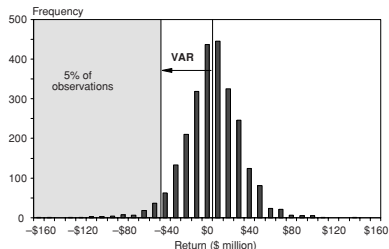
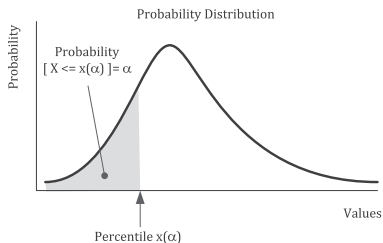
A percentile of a target random variable is a threshold value of this variable that, depending on the issue, is an upper bound not exceeded with a predetermined probability, or, alternatively, is a lower bound such that the probability of the target variable being higher is predefined.

Value-at-Risk

Percentile and confidence level

- N.B. Risk addresses losses or negative earnings $\Rightarrow$ we look for upper bounds of losses or for lower bounds of earnings defined through percentiles of the distribution of Profits and losses (P&L, returns)
- A percentile is always defined as the probability (α) of a variable being lower or equal to an upper bound that we name $VaR(\alpha)$
- Here α = coverage rate ($1 - \alpha$ = confidence interval) and L the loss

$$P(P\&L < VaR) = \alpha$$



shows the α percentile of X , $x(\alpha)$.

The probability that X is lower than this upper bound is the area under the curve, or the cumulative probability density of X at $(x(\alpha))$.

The probability that X exceeds $x(\alpha)$, considered as a lower bound, is $1 - \alpha$, and is the remaining area on the right-hand side of $x(\alpha)$.

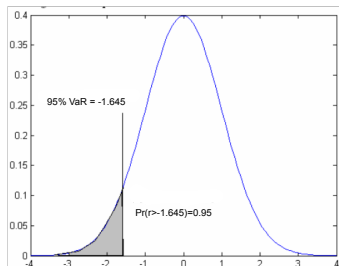
VAR then is the cutoff loss such that the probability of experiencing a greater loss is less than α percent.

Value-at-Risk

Statistical Definition

- A VaR defined for a coverage rate $\alpha\%$ corresponds to the α quantile of the P&L distribution associated with holding an asset or a portfolio over a fixed period
- typical values for the confidence level $1 - \alpha$: 0.95, 0.99 or 0.999

e.g. VaR p.d.f. under the Normal distribution



Value-at-Risk computation

Elements required to compute the VaR

- the P&L distribution for the asset or the portfolio, which accounts for the allocation in terms of risky assets (related to risk factors)
- the coverage rate (1- the confidence level), the probability that the loss is lower than the VaR, e.g. 5%
- the holding period (H days for which the loss is computed)

Hence, if the return distribution is known, the VaR can be immediately derived as the α quantile of the P&L distribution $VaR_t(\alpha) = F_{R_t}^{-1}(\alpha)$

Otherwise, estimate the parameters of the underlying loss distribution, and evaluate the inverse cdf at required quantile

Basel II (2007): bank regulators require banks to calculate VaR for market risk with $H = 10$ and $1 - \alpha = 0.99$ based on at least one year of data.

Basel capital is based on a 10-day VaR, which is designed to account for the potential inability of banks to hedge or trade for as long as ten days because of market illiquidity.

In practice, banks have avoided the need to use 10-day returns by multiplying one-day VaR by $\sqrt{10}$, which is permitted by the Basel capital rules.

VaR and Economic Capital

- Economic capital is the capital required to absorb losses in excess of expected loss due to current risks and is defined by the VaR at a given confidence level
- For preserving solvency of financial institutions in adverse conditions, capital should be capable of absorbing such excess losses
- The VaR represents the capital in excess of expected loss necessary for absorbing unexpected deviations from average losses with a given probability
- Using $\alpha = 1\%$ for VaR would imply that capital is not high enough to absorb losses two or three times a year (250 business days in a year)
- For market risk, regulators allow the use of internal VaR models using the 1% confidence level, but they impose a capital higher than VaR by a multiple currently between 3 and 4

VaR and Economic Capital

Under Basel 2.5 framework the required capital is

$$\mathcal{K}_t = \mathcal{K}_t^{VaR} + \mathcal{K}_t^{SVaR} + \mathcal{K}_t^{SRC} + \mathcal{K}_t^{IRC} + \mathcal{K}_t^{CRM}$$

- $\mathcal{K}_t^{VaR}$ is the VaR capital
- $\mathcal{K}_t^{SVaR}$ is the Stressed VaR (captures stressed market conditions)
- $\mathcal{K}_t^{SRC}$ is the solvency capital requirement (Basel II)
- $\mathcal{K}_t^{IRC}$ is the incremental risk charge (IRC), which measures the impact of rating migrations and defaults
- $\mathcal{K}_t^{CRM}$ is the comprehensive risk measure (CRM), which corresponds to a supplementary capital charge for credit exotic trading portfolios

Definition

The stressed VaR has the same characteristics than the traditional VaR (99% confidence level and 10-day holding period), but the model inputs are calibrated to historical data from a continuous 12-month period of significant financial stress relevant to the bank's portfolio.

Value-at-Risk computation

Single asset or portfolio ?

In the second case there are 2 ways:

- Consider the return of the portfolio as that of an asset and compute the VaR on this aggregated return
 - Simple to implement, but neglecting the covariances between asset returns means not accounting for any possible risk diversification
- Account for the correlations between assets when computing the VaR (multivariate approach)

Value-at-Risk estimation methods

- 1 Non-parametric methods (Historical Simulation, Weighted Historical Simulation, Filtered historical Simulation)
- 2 Semi-parametric methods: CAViaR (Engle and Manganelli , 2004), extreme value theory (McNeil and Frey , 2000)
- 3 Parametric methods (ARCH, univariate GARCH, multivariate GARCH, RiskMetrics)

Remark: In the rest of the slides, the VaR will be expressed as a positive number, like it is done in the literature

Engle, R. F., and Manganelli, S. (2004). CAViaR: Conditional autoregressive value at risk by regression quantiles. *Journal of Business & Economic Statistics* 22, 367-381.

McNeil, A., and Frey, R. (2000). Estimation of Tail Related Risk Measure for Heteroscedastic Financial Time Series: An Extreme Value Approach. *Journal of Empirical Finance* 7, 271-300.

directly estimate the associated quantile of historical data : **order statistic approach**

mix of the two

estimate model for underlying loss distribution, and evaluate inverse cdf at required quantile

modelling the loss distribution inevitably entails model risk, which is concerned with possibly misleading results due to model misspecifications

non-param: Historical Simulation (HS) 2 Bootstrapped Historical Simulation 3 Simulation Historique et Estimation Non Paramétrique de Densité 4 Weighted Historical Simulation (WHS) ou Hybrid Method 5 Filtered Historical Simulation (FHS)

CAViaR conditional autoregressive specification for VaR - based on quantile regression

- $VaR_t = \beta_0 + \sum \beta_i VaR_{t-i} + fct of past returns$ (it measures how much the VaR should change based on the new information in returns / absolute returns / + - returns)

- estimated using Koenker and Bassett's (1978) regression quantile framework

EVT method : based on the generalized Pareto distribution GPD fitted by MLE for eg in the reference above

Value-at-Risk estimation

1. Nonparametric methods: Historical Simulation

- Very simple, the VaR is the empirical quantile of the past returns
- Example: If we have a sample of 1000 observations of past returns, the VaR for a 95% confidence interval is the return value corresponding to the 50st largest loss
- If the order statistic is not an integer, one can take the interpolation scheme

no parametric distribution is imposed a priori

Value-at-Risk estimation : 1. HS

- Let $R_1, R_2, \dots, R_T$ be a sequence of returns observed at time $t = 1 \dots T$
- At each date, the return R_t is a random variable drawn from a certain distribution for which the associated $VaR_t(\alpha)$ can be deduced
- Under this hypothesis of i.i.d.-ness of the returns, we obtain a sample of T realizations of the random variable that follow the same distribution and hence admit the same VaR.

Value-at-Risk estimation : 1. HS

Definition

Under the previous hypotheses, a convergent estimator of the VaR for a coverage rate $\alpha\%$ is defined by the α empirical quantile of the historical realizations of the returns, denoted by $r_1, r_2, \dots, r_T$

$$\widehat{VaR}(\alpha) = percentile(\{r_j\}_{j=1}^T, 100\alpha)$$

$$\widehat{VaR}(\alpha) \xrightarrow{p} VaR(\alpha)$$

Value-at-Risk estimation : 1. HS

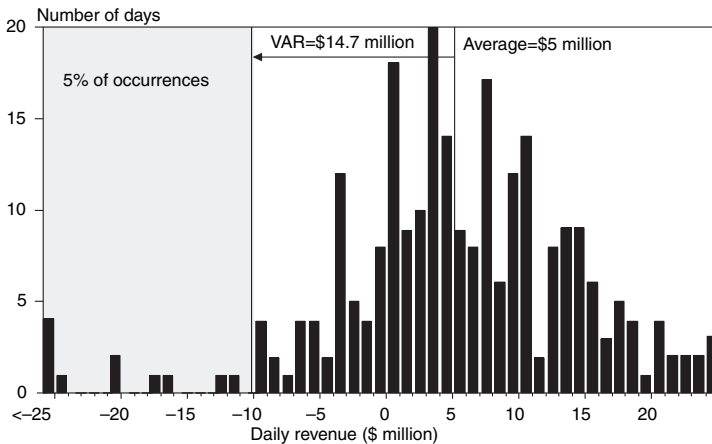
- Let us consider the daily returns of J.P. MORGAN in 1994 (a total of 254 observations)
- We sort the observations in ascending order $\{r_1, \dots, r_{254}\}$
- Compute the 5% VaR-HS as the $254 \times 5\% = 12.7$ th observation
- We have 11 observations to the left of -\$10 million and 15 to the left of -\$9 million
- Interpolating, we find

$$\widehat{VaR}_t(5\%) = -\$9.6\text{million}$$

The VAR of daily revenues, measured relative to the mean, is $VAR_m = E(P\&L) - VaR = \$5.1 - (-\$9.6) = \14.7 million

Value-at-Risk estimation : 1. HS

Computation of nonparametric VAR.



J.P. Morgan's distribution of daily revenues in 1994. The graph shows how to compute nonparametric VAR.

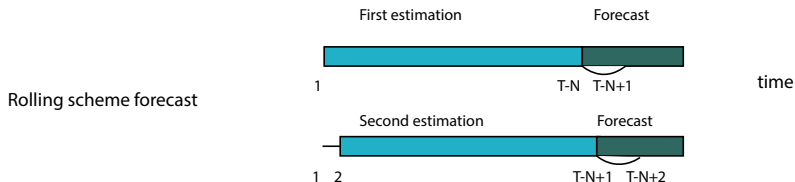
From this graph, the average revenue is about \$5.1 million. There is a total of 254 observations; therefore, we would like to find W^* such that the number of observations to its left is $254 \times 5 \text{ percent} = 12.7$.

We have 11 observations to the left of -\$10 million and 15 to the left of -\$9 million. Interpolating, we find $W^* = -\$9.6$ million. The VAR of daily revenues, measured relative to the mean, is $\text{VAR} = E(W) - W^* = \$5.1 - (-\$9.6) = \14.7 million.

If one wishes to measure VAR in terms of absolute dollar loss, VAR then is \$9.6 million. Finally, it is useful to describe the average of losses beyond VAR, which is \$20 million here. Adding the mean, we find an expected tail loss (ETL) of \$25 million.

Value-at-Risk estimation : 1. HS

How to obtain a sequence of VaR-HS forecasts ?



Pros. Easy to implement. No statistical estimation of the distribution of X necessary.

Cons. It may be difficult to collect sufficient quantities of relevant, synchronized data for all risk factors. Historical data may not contain examples of extreme scenarios.

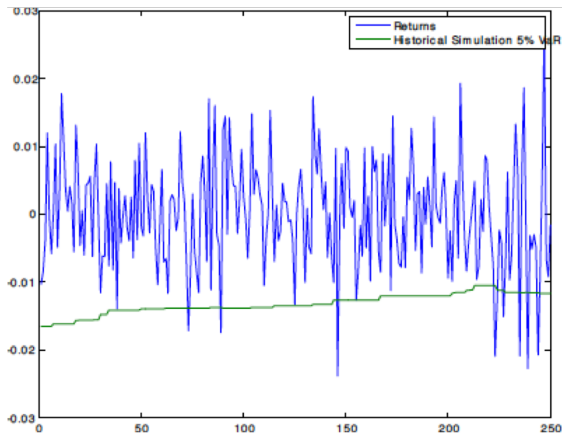
Monte Carlo method: Instead we make inference about L using Monte Carlo methods, which involves simulation of new risk factor data.

Pros. Very general. No restriction in our choice of distribution for X_{t+1} .

Cons. Can be very time consuming if loss operator is difficult to evaluate, which depends on size and complexity of portfolio.

Value-at-Risk estimation : 1. Rolling HS

⇒ VaR-HS forecasts are relatively invariant to changes in the economic environment, i.e. VaR-HS forecasts are ‘flat’ or ‘relatively flat’ (in the case of rolling estimates).



Fixed window size T_e

Plus la taille de la fenêtre T_e est petite, plus les prévisions de VaR seront volatiles
Plus la taille de la fenêtre T_e est grande, plus la VaR prévue convergera vers la VaR non conditionnelle et sera par conséquent quasi "constante" dans le temps.

bootstrapped HS: take average of VaR on bootstrapped samples - from each window for rolling forecasts

weighted HS: do not give same weight to the recent and old observations / or depending on volatility level, or correlation level

Value-at-Risk estimation

Parametric methods: steps

- 1 Stipulate a parametric distribution for the P&L (normal, Student, GED, ...)
- 2 Find the estimators of the parameters of that distribution (estimation)
- 3 Compute the appropriate quantile

Value-at-Risk estimation : Parametric methods (hypothesis H1)

H1 Assume normally distributed losses with mean μ and variance σ^2 , $R_t \sim N(\mu, \sigma^2)$

Definition

VaR under the normality of the P&L hypothesis for a coverage rate of $\alpha\%$ is

$$VaR_t(\alpha) = -\mu - \sigma\Phi^{-1}(\alpha),$$

with Φ the standard normal cumulative distribution function (Recall VaR is taken as a positive number here)

Value-at-Risk estimation : Parametric methods (hypothesis H1)

Proof

$$\begin{aligned}\Pr[R_t < -VaR_t(\alpha)] &= \alpha \\ \Leftrightarrow \Pr\left[\frac{R_t - \mu}{\sigma} < \frac{-VaR_t(\alpha) - \mu}{\sigma}\right] &= \alpha\end{aligned}$$

where the standardized variable follows a standard normal distribution by H1

Therefore,

$$\begin{aligned}\Phi\left(\frac{-VaR_t(\alpha) - \mu}{\sigma}\right) &= \alpha \\ \Rightarrow \frac{-VaR_t(\alpha) - \mu}{\sigma} &= \Phi^{-1}(\alpha)\end{aligned}$$

Value-at-Risk estimation : Parametric methods (hypothesis H1)

Numerical example 1

- $R_t \sim N(0.01, 1.5^2)$

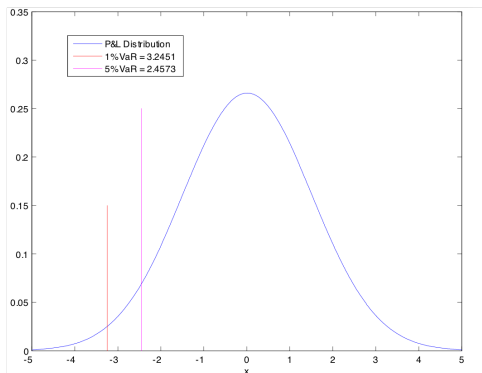
⇒ the 1% and 5% VaR are

- $VaR_t(0.01) = -0.01 - 1.5\Phi^{-1}(0.01) = 3.479$

- $VaR_t(0.05) = -0.01 - 1.5\Phi^{-1}(0.05) = 2.457$

Value-at-Risk estimation : Parametric methods (hypothesis H1)

Numerical example



In practice, the assumption of **normally distributed returns** usually can be rejected both for loss distributions associated with credit and operational risk, as well as for loss distributions associated with market risk at high levels of confidence

Value-at-Risk estimation : Parametric methods (hypothesis H2)

H2 We suppose that the P&L follows a Student distribution with ν degrees of freedom

$$R_t \sim t(\nu)$$

Definition

VaR under the Student distribution for an $\alpha\%$ coverage rate is

$$VaR_t(\alpha) = G(\alpha; \nu)^{-1}$$

Where $G(\alpha; \nu)^{-1}$ represents the cumulative distribution function for a Student distribution with ν degrees of freedom

Value-at-Risk estimation : Parametric methods (hypothesis H2)

Numerical example

- $R_t \sim t(5)$

⇒ the 1% and 5% VaR are

- $VaR_t(0.01) = G^{-1}(0.01, 5) = 4.032$

- $VaR_t(0.05) = G^{-1}(0.05, 5) = 2.570$

Value-at-Risk estimation: Parametric methods (conditional distribution)

- We generally work with the conditional distribution of P&L instead of the marginal one
- ⇒ P&L Density function conditional on the set of information available at time t , Ω_t , with r_t a realization (observation) from the sample of return process R_t

Definition

Conditional VaR for a coverage rate $\alpha\%$ conditional on Ω_t corresponds to the α quantile of the conditional distribution of the P&L

$$VaR_t(\alpha|\Omega_t) = F_{R_t}^{-1}(\alpha|\Omega_t)$$

where $F_{R_t}(\alpha|\Omega_t)$ is the conditional CDF of returns.

Value-at-Risk estimation: Parametric methods (conditional distribution)

Example

- Suppose the P&L at time $t + 1$ denoted by R_{t+1} are normally distributed
- The conditional distribution of the P&L is

$$R_{t+1}|\Omega_t \sim N(\hat{\mu}_{t+1|t}, \hat{\sigma}_{t+1|t}^2),$$

with

$$\hat{\mu}_{t+1|t} = \mathbb{E}(R_{t+1}|\Omega_t)$$

$$\hat{\sigma}_{t+1|t}^2 = \mathbb{V}(R_{t+1}|\Omega_t)$$

- E.g. obtained by estimating a GARCH model
- The (negative) VaR forecast at $t + 1$ for a coverage rate α is

$$VaR_{t+1|t}(\alpha) = -\hat{\mu}_{t+1|t} - \hat{\sigma}_{t+1|t}\Phi^{-1}(\alpha)$$

Value-at-Risk advantages and inconvenients

Advantages of the VaR as a risk measure

- Single number that indicates the extent of risk in a given portfolio and is measured in either price or as a percentage, making understanding VaR easy
- Simple to compute and interpret

Value-at-Risk advantages and inconvenients

Inconvenients of the VaR as a risk measure

General ones

- Subjected to model risk (e.g. specification error in the P&L distribution)
- Subjected to implementation risk

Specific ones

- Does not provide any information about the losses beyond the VaR (worst possible and their distribution)
- Misleading; different methods can lead to different results
- Can lead agents to take more risk
- Can be difficult to calculate the VaR for large portfolios

While VaR is useful for predicting the risks facing an investment, it can be misleading. One critique is that different methods give different results: you might get a gloomy forecast with the historical method while Monte Carlo Simulations are relatively optimistic. It can also be difficult to calculate the VaR for large portfolios: you can't simply calculate the VaR for each asset, since many of those assets will be correlated. Finally, any VaR calculation is only as good as the data and assumptions that go into it

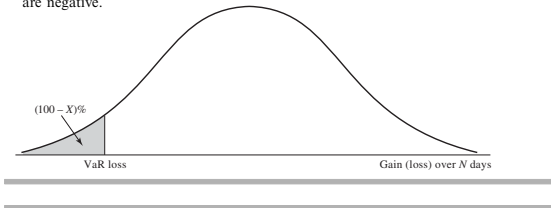
VAR does not say anything about the distribution of losses in its left tail.

VAR is measured with some error. The VAR number itself is subject to normal sampling variation.

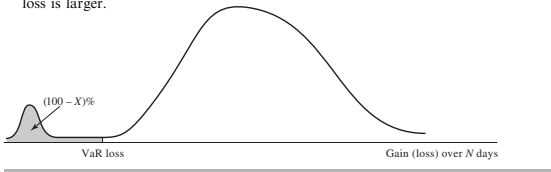
An advantage of the parametric approach is that VAR is more precisely estimated than with historical simulation. This is because the standard deviation estimator uses all the observations in the sample.

Value-at-Risk advantages and inconvenients

Calculation of VaR from the probability distribution of the change in the portfolio value; confidence level is $X\%$. Gains in portfolio value are positive; losses are negative.



Alternative situation to Figure 22.1. VaR is the same, but the potential loss is larger.



Two portfolios can have the same VaR but different extreme risk levels (no information about the size of extreme loss is given by the VaR)

Note also that we focus on the univariate case here

Given only information about VaR^A of random variable A and VaR^B of random variable B, there is in general no sufficient information to calculate VaR for a function of both: $VaR^{f(A;B)} \neq g(VaR^A; VaR^B)$

- In such cases, in order to calculate $VaR^{f(A;B)}$, we have to derive the distribution of $f(A;B)$ first despite the marginal distributions of the constituting parts, the transformed distribution under f is affected by the way that the margins are related with each other: the dependence structure between individual assets is crucial to the determination of $VaR^{f(A;B)}$

- As multi-period returns can be calculated as simple sum of sub-period returns in the logarithmic case, we aim to model $VaR^{f(A;B)} = VaR^{A+B}$ even though our object of interest relates to a simple sum of random variables, easy analytical solutions apply only in the very restricted cases where summation preserves the distribution: A, B and $A+B$ have to be of the same distribution this property is fulfilled for the case of jointly normally distributed random variables

- while portfolio returns can be calculated as weighted sum of individual assets for discrete returns, such an easy relation does not exist for the case of logarithmic returns

What is a good risk measure?

The Coherent risk measure theory of Artzner et al. 1997, 1999

Artzner, P., Delbaen, F., Eber, J.M., Heath, D., 1997. Thinking coherently. Risk 10 (11), 68-71.

Artzner, P., Delbaen, F., Eber, J.M., Heath, D., 1999. Coherent measures of risk. Mathematical Finance 9(3), 203-228.

What is a good risk measure?

The Coherent risk measure theory of Artzner et al. 1997, 1999

Definition

Let X and Y be two $P\&L$ distributions associated with two portfolios and $\rho(\cdot)$ a risk measure over a given horizon

The risk measure $\rho(\cdot)$ is said to be coherent if and only if it satisfies the following axioms:

- | | |
|--------------------------|---|
| 1 Monotonicity | $Y \succ X \Rightarrow \rho(Y) \leq \rho(X)$ |
| 2 Subadditivity | $\rho(X + Y) \leq \rho(X) + \rho(Y)$ |
| 3 Homogeneity | $\rho(hX) = h\rho(X)$ for $h > 0$ |
| 4 Translation invariance | $\rho(X + n) = \rho(X) + n, \forall n \in \mathbb{R}$ |

What is a good risk measure?

The Coherent risk measure theory of Artzner et al. 1997, 1999

Remark The subadditivity axiom is the most important

Theorem VaR is not a coherent risk measure;
In particular, it does not fulfil the subadditivity axiom (Note however that Value at Risk is coherent for Gaussian random variables)

- There are other risk measures, e.g. Expected Shortfall, that are subadditive in all cases

The standard deviation, however, is subadditive: The SD of a portfolio must be smaller than, or at worst equal to, the sum of the SDs of subportfolios. As a result, VAR computed from the standard deviation is subadditive as well.

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 - Backtesting ES
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Expected Shortfall

Definition

The Expected Shortfall associated with an $\alpha\%$ coverage rate denotes the conditional expected loss (L) given that the realized loss is equal to or exceeds the corresponding value of the $Var(\alpha)$

$$ES(\alpha) = \mathbb{E}(L | L \geq Var(\alpha))$$

Otherwise said, it corresponds to the average of the $\alpha\%$ largest expected losses

$$ES(\alpha) = -\frac{1}{\alpha} \int_0^\alpha F_{R_t}^{-1}(p) dp$$

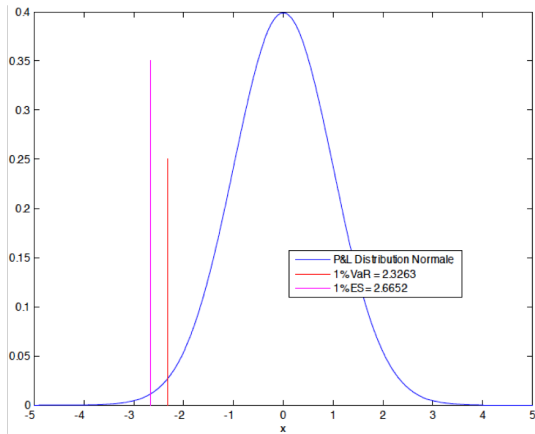
where $F_{R_t}^{-1}$ stands for the cumulative distribution function of the returns series

It hence coincides with the Tail Value at Risk (conditional VaR)

Expected Shortfall computation

Example In the case of a normal distribution, the ES associated with an $\alpha\%$ coverage rate is

$$ES_t(\alpha) = -\mu - \sigma \frac{\phi(\Phi^{-1}(\alpha))}{1 - \alpha}$$



Expected Shortfall computation

- The value-at-risk and the expected shortfall are both a standard deviation-based risk measure. They coincide when the scaling parameters are equal:

$$c_{VaR} = \Phi^{-1}(\alpha_{VaR})$$

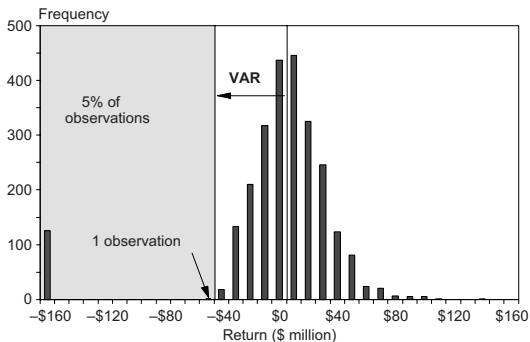
$$c_{ES} = \phi(\Phi^{-1}(\alpha_{ES})) / (1 - \alpha_{ES})$$

Table: Scaling factors c_{VaR} and c_{ES}

α (in %)	95.0	96.0	97.0	97.5	98.0	98.5	99.0	99.5
c_{VaR}	1.64	1.75	1.88	1.96	2.05	2.17	2.33	2.58
c_{ES}	2.06	2.15	2.27	2.34	2.42	2.52	2.67	2.89

Expected Shortfall computation

- In general the underlying distribution is not known
- Two estimation methods of ES
 - Directly estimate the mean of all values greater than the associated quantile of historical data
 - Estimate a model for underlying loss distribution, and calculate expectation of conditional loss distribution



ES_MSE.pdf. figure : high mean squared errors (mse) for Expected Shortfall at high confidence levels

In addition, CVAR is better justified than VAR in terms of decision theory.

Suppose an investor has to choose between two portfolios A and B with different distributions. Decision rules can be based on various definitions of stochastic dominance. One example is first-order stochastic dominance (FSD). This requires that the cumulative distribution function for A be systematically lower than that for B. So, B has a higher probability of a bad outcome. This is a very strict rule, however. Another example is second-order stochastic dominance (SSD). Portfolio A would dominate B if it has higher mean and lower risk. This is a more realistic rule than the first.

Using CVAR as a risk measure is consistent with **SSD**, whereas VAR requires **FSD**, which is less realistic.

VaR and ES meaning

- You have invested 500,000 euro in an investment fund. The manager of the fund tells you that the 99% Value-at-Risk for a time horizon of one year amounts to 5% of the portfolio value. Explain the information conveyed by this statement.

VaR and ES meaning

- For a continuous loss distribution we have $\mathbb{P}(L \geq VaR_\alpha) = 1 - \alpha$
- Compute $VaR_\alpha = 0.05 \times 500,000 = 25,000$
- Plugging into the formula yields $\mathbb{P}(L \geq 25,000) = 0.1$ interpretable as “with probability 1% you will loose 25,000 euro or more”
- A capital cushion corresponding to $VaR(0.01) = 25000$ is sufficient in 99% of the cases

VaR and ES meaning

- The funds manager corrects himself, instead of the VaR, it is the ES that amounts to 5% of the portfolio value.

How does this statement have to be interpreted? Which of the two cases implies a riskier portfolio?

VaR and ES meaning

- If one of the worst 1% years occurs in the next period, the expected loss in this year will amount to $ES = 25,000$ euro
- At the same time, since the ES is always larger the VaR (in absolute values), the first statement implies that $ES(0.01) \geq 25,000$ euro
 $\Rightarrow$ the first statement implies a riskier portfolio

Multi-period risk measures computation

- Basel Committee fixed a time-horizon of 10 days for VaR & ES
- Assumption of independence of returns over time + **normally distributed** returns
 - moments of multi-period returns can be derived from moments of one-period returns: square-root-of-time scaling for standard deviation
 - multi-period returns follow normal distribution: VaR and ES can be derived according to square-root-of-time scaling
- Assumption of independence of returns over time + **arbitrary return distribution**
 - moments of multi-period returns can be derived from moments of one-period returns: square-root-of-time scaling for standard deviation
 - multi-period return distribution is unknown: for some important risk measures like VaR or ES no analytical solution exists



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Backtesting

- Value-at-risk models are only useful insofar as they predict risk reasonably well $\Rightarrow$ Model validation
- Distinguish
 - **Stress tests** serve for assessing what would happen under stressed conditions
 - **Backtests** serve for checking that the risk measures effectively capture actual observed variations
 - **Benchmarking** compares outputs of various models to alternate models serving as benchmarks

A back test compares “before the fact” measures with “after the fact” measures. consistency with reality.

A stress test, in its simplest form, is a sensitivity analysis with a focus on critical variables.

Benchmarking is essentially a comparison of models.

— next slide:

VaR procyclical : should move in line with current conditions for being relevant. Using longer windows or different methodologies for estimating volatility would result in different estimates of VaR because it is highly sensitive to the behavior of volatility. Using a moving average of volatility or a GARCH model would result in very different VaRs. The choice of methodology for modeling volatility depends on the relevance of the calculations. Using longer windows of observations might make it smoother, except that when the window moves through time, some historical shocks might disappear, resulting in abrupt movements of VaR. Whatever the method, the VaR should pass simple back tests.

—

it is one of the reasons that bank regulators allow banks to use their internal risk measures to determine the amount of regulatory capital required to support their trading portfolios

Backtesting

Value-at-risk models are only useful insofar as they predict risk reasonably well $\Rightarrow$ Model validation

Definition

Backtesting is a formal statistical framework that consists in verifying that the actual losses are in line with the projected losses.

This implies systematically comparing the history of risk forecasts (daily VaR forecasts) with next day's realized portfolio returns. (Jorion, 2007)

■ Why use backtesting?

- Financial regulation
- Risk-managers
- Variety of forecasting models yielding possibly different results

Backtesting: role

- Identifying the weaknesses of risk forecasting methods $\Rightarrow$ avenues for improvement
- Models that perform poorly during backtesting should question
 - 1 model assumptions
 - 2 parameter estimates
- Backtesting can prevent underestimation and overestimation of risk

Backtesting vs. Forecast evaluation test

- Traditionally, we compare the forecast (ex-ante) with the ex-post realization
- The test hence relies on a criterion based on the distance between the true and forecasted values (loss function), e.g. Mean-Square Error, or information criteria (AIC, BIC).

Remark No ex-post observation of the VaR is available and no satisfactory proxy either!

- ⇒ compare the actual portfolio variations of value, given historical data, with the modeled VaR

Backtesting: VaR violation

- When the model is perfectly calibrated, the number of returns falling above the VaR should be in line with the confidence level
- The number of exceedences is also known as the number of **exceptions**

Definition

A VaR violation (hit, exception) is a situation where the observed loss at time t is larger than the forecasted VaR

The hit function (hit variable) is a dummy variable $I_t(\alpha)$ associated with the observation of an $\alpha\%$ VaR violation at time t

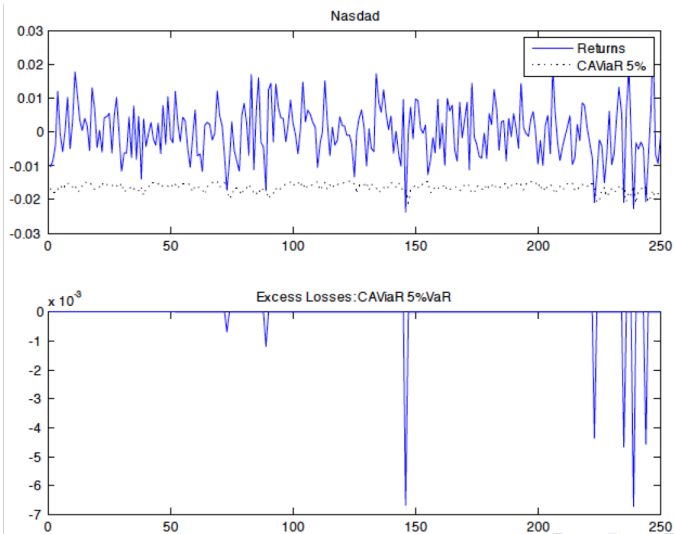
$$I_t(\alpha) = \begin{cases} 1, & \text{if } R_t < VaR_{t|t-1}(\alpha) \\ 0 & \text{otherwise} \end{cases}$$

Note that the VaR is defined as the true, negative quantile here

- With too many exceptions, the model underestimates risk

too many exceptions: This is a major problem because too little capital may be allocated to risk-taking units; penalties also may be imposed by the regulator. Too few exceptions are also a problem because they lead to excess, or inefficient, allocation of capital across units.

Backtesting: VaR violation



Backtesting in practice

- VaR forecasts are compared with the actual outcome
- Daily returns over T periods are known, say from 2017 to 2021 and assume 250 trading days/year
- Use a rolling window procedure with $N = 500$ (estimation window, i.e. observation used for forecasting risk)
- The testing window is therefore of XXX days (over how many days is risk forecasted ?)

$250 \times 5 - 500 = 750$ oos 1 period ahead VaR forecasts

Backtesting in practice

- The main tools used in backtesting are **violation ratios**

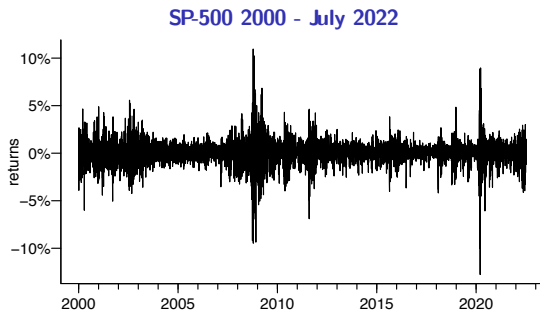
$$VR = \frac{\text{Observed } N^{\circ} \text{ of violations}}{\text{Expected } N^{\circ} \text{ of violations}}$$

- The **observed** number of VaR violations are compared with the **expected**
- If the violation ratio is greater than one the VaR model **underforecasts risk**
- If smaller than one the model **overforecasts risk**

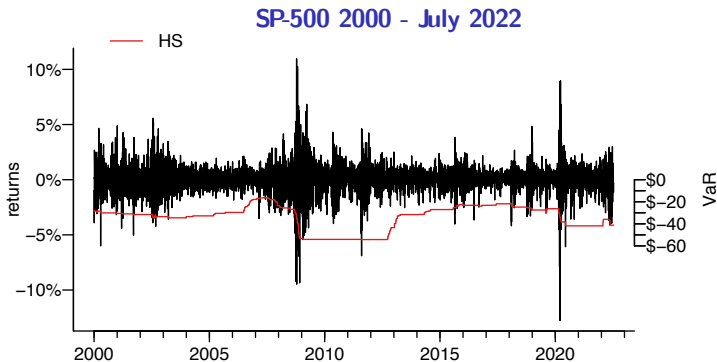
- $250 \times 5 - 500 = 750$ oos 1 period ahead VaR forecasts
 - Expected N^o of violations = theoretical proba of occurrence α times the OOS sample size
 - VaR violations are infrequent events: With a 1% VaR, a violation is expected once every 100 days, so that 2.5 violations are expected per year
 - So the actual sample size of violations is quite small, Causing difficulties for statistical inference
 - Need at least 10 violations for reliable statistical analysis, or 4 years of data, Preferably more
 - VR=1 is expected, A useful rule of thumb is
- If VR in $[0.8, 1.2]$ the model is good
- If VR in $[0.5, 0.8]$ or VR in $[1.2, 1.5]$ the model is acceptable
- If VR in $[0.3, 0.5]$ or VR in $[1.5, 2]$ the model is bad
- If VR < 0.3 or VR > 2 the model is useless
- Plot the actual returns and VaR together And then do a statistical test

Backtesting: Visual analysis (cf. Danielsson 2022)

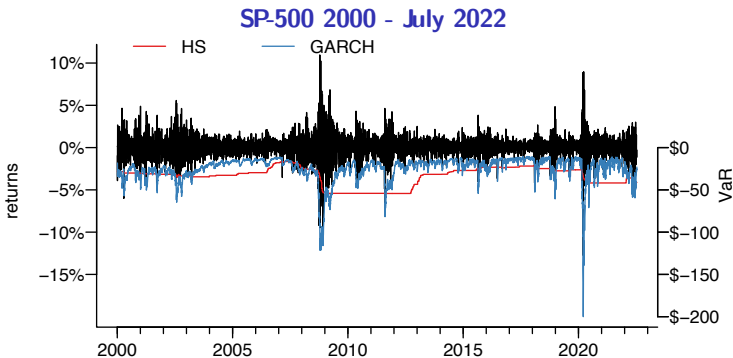
- Take a sample of S&P500 and plot the actual returns and VaR together
- Discuss interesting regimes



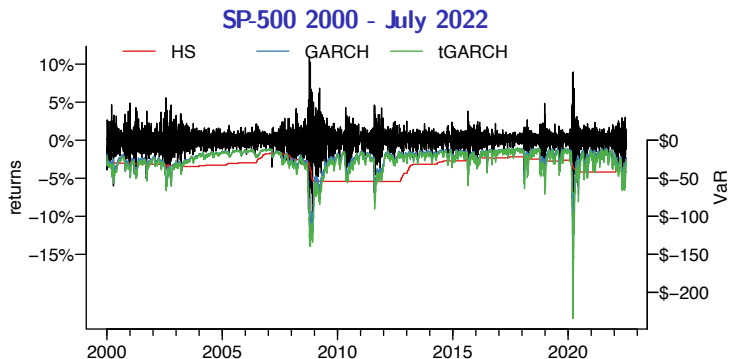
Backtesting: Visual analysis (cf. Danielsson 2022)



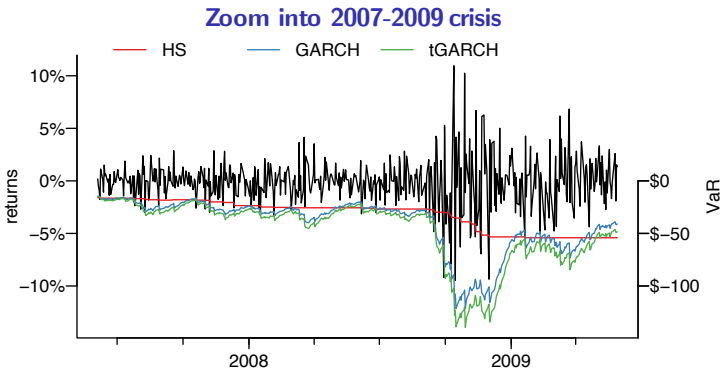
Backtesting: Visual analysis (cf. Danielsson 2022)



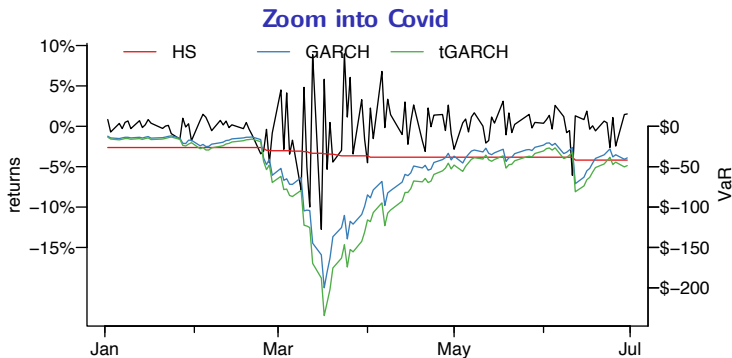
Backtesting: Visual analysis (cf. Danielsson 2022)



Backtesting: Visual analysis (cf. Danielsson 2022)



Backtesting: Visual analysis (cf. Danielsson 2022)



Backtesting methodology

Backtesting VaR-calculations based on the assumption of independent normally distributed losses generally may lead to two patterns

- percentage frequencies of VaR-exceedances are higher than the confidence levels specified: normal distribution assigns a too small probability to large losses
 - VaR-exceedances occur in clusters: given an exceedance of VaR today, the likelihood of an additional exceedance in the following days is larger than average
- ⇒ VaR evaluation is therefore based on **statistical tests** (not simple criteria) for the two main hypotheses that the VaR violation process must fulfill:
- Unconditional coverage hypothesis (test the N^α of violations)
 - Independence hypothesis (test clustering)

Backtesting VaR methods

- Unconditional coverage test : Kupiec (1995); Traffic light Basel II (BSBC 1996)
- Conditional coverage test : Christoffersen (1998)
- Martingale difference hypothesis test : Berkowitz (2005); Hurlin and Tokpavi (2007)
- Regression model tests : Engle and Manganelli (2004); Dumitrescu, Hurlin and Pham (2012)
- Duration based tests : Christofffersen and Pelletier (2004); Candelon et al. (2008)
- Density forecasts tests : Diebold et al (1998); Berkowitz (2001)
- Evaluation of backtesting procedures : Hurlin and Tokpavi (2008)

Backtesting ES methods

- Harder than for the VaR because we are testing an **expectation** rather than a **single quantile**
- We know if VaR is violated, but cannot know that for ES
- Some tests:
 - Acerbi and Szekely (2014)
 - Nolde and Ziegel (2017)
 - Du and Escanciano (2017)
 - Bayer and Dimitriadis (2022)

several tests require the **whole or tail distribution of the returns** or equivalently the cumulative violation process (Kerkhof and Melenberg, 2004; Wong, 2008; Graham and Pal, 2014; Acerbi and Szekely, 2014; Du and Escanciano, 2017; Loser, Wied, and Ziggel, 2018; Costanzino and Curran, 2018),
= **multiple quantiles at different levels** (Emmer, Kratz, and Tasche, 2015; Costanzino and Curran, 2015; Kratz, Lok, and McNeil, 2018; Couperier and Leymarie, 2019),
= **the VaR and the volatility** (McNeil and Frey, 2000; Nolde and Ziegel, 2017; Righi and Ceretta, 2013, 2015),
= **or the VaR** (McNeil and Frey, 2000; Nolde and Ziegel, 2017)
in addition to the ES forecasts. See Section S.1.2 in the Supplementary Appendix for an overview over the existing backtesting approaches.
Du and Escanciano (2017) Rely on the MDS framework of Berkowitz (2005, 2011) : Cumulative violations (integral of violations over the coverage level in the left tail) should form a MDS if the risk model is appropriate

Backtesting: Possible issues

- Backtests are very sensitive to data contamination (Fresard, Perignon and Wilhelmsson, 2011)
- Backtesting assumes that there have been no structural breaks in the data throughout the testing period
- Backtesting is only statistically valid if we have no ex-ante knowledge of the data in the testing window
- In practice, banks appear to systematically overestimate their VaR during tranquil periods and to underestimate it during turmoil
- To come up with scenarios that are not well represented in recent historical data but are nonetheless possible and detrimental to portfolio performance, banks are now required to calculate stressed VaR

-over the period 2005–2008, fewer than 6% of the largest commercial banks in the world evaluated their VaR models using the appropriate uncontaminated data (by extraneous profits such as intraday revenues, fees, commissions, net interest income, and revenues from market making or underwriting activities)

under/overestim: due, at least to some extent, to the fact that, typically, VaR measures are calibrated using historical data. Following a period of calm in financial markets, the VaR estimates and the accompanying CR may decline to low levels, but might then underestimate the risk during a future period of stress.

But financial markets are continually evolving,
new technologies, assets, markets and institutions affect the statistical properties of market prices

Unlikely that the statistical properties of market data in the 1990s are the same as today,

implying that a risk model that worked well then might not work well today

=====

If we iterate the process, continually refining the risk model with the same test data and thus learning about the events in the testing window,
the model will be fitted to those particular outcomes,
violating underlying statistical assumptions

So the actual confidence bounds are wider than suggested by the testing

===== Stressed VaR

While there are several ways to do that, here is a really simple approach -Suppose we have a sample $1, \dots, WE, \dots, T$ - We have a VaR_{t+1} - The stressed VaR is -
 $VaR_{t+1}^S = \max VaR_i, i = \{WE + 1, \dots, T + 1\}$

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Backtesting VaR: Christoffersen (1998) strategy

Definition (Unconditional coverage)

The Unconditional coverage hypothesis is satisfied when the probability to observe an ex-post excess loss relative to the ex-ante VaR forecast is precisely equal to the coverage rate α

$$\Pr[I_t(\alpha) = 1] = \mathbb{E}[I_t(\alpha)] = \alpha$$

■ Example:

In the case of a 1%VaR used as a reference over a period of 500 observations, the expected value of the number of violations should equal 5.

Backtesting VaR: Christoffersen's (1998) general strategy

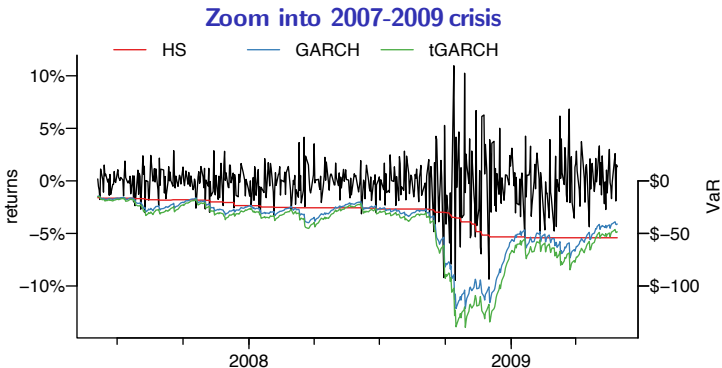
Definition (Independence)

The Independence hypothesis is satisfied when VaR violations observed at two different dates, for the same coverage rate, are independently distributed. Formally, $I_t(\alpha)$ associated with a VaR violation at time t for a coverage rate α is independent of $I_{t \pm k}(\alpha)$, $\forall k \neq 0$

Corollary

Under the independence hypothesis there are no clusters of violations

Backtesting VaR: Christoffersen's (1998) general strategy



Backtesting VaR: Traffic Light approach (BCBS 1998)

The Basel Penalty Zones

Zone	Number of Exceptions	Increase in k
Green	0 to 4	0.00
Yellow	5	0.40
	6	0.50
	7	0.65
	8	0.75
	9	0.85
Red	10+	1.00

- Green zone : model is likely to be correct.
- Yellow zone : Outcomes in this range are plausible for both accurate and inaccurate models
- Red zone : Automatic presumption that a problem exists with a bank's model

Within the “yellow” zone, the penalty is up to the supervisor, depending on the reason for the exception. The Basel Committee uses the following categories:

- Basic integrity of the model. The deviation occurred because the positions were reported incorrectly or because of an error in the program code.
- Model accuracy could be improved. The deviation occurred because the model does not measure risk with enough precision (e.g., has too few maturity buckets).
- Intraday trading. Positions changed during the day.
- Bad luck. Markets were particularly volatile or correlations changed.

For the first two reasons, the penalty “should” apply.

With the third reason, a penalty “should be considered.



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Backtesting ES

Bayer and Dimitriadis (2022)

- Require solely ES forecasts for backtest (and realized -OOS- returns)
- Based on the idea of Mincer Zarnowitz (1969) regression-based forecast evaluation

$$ES_t(R_t|\mathcal{I}_{t-1}) = \gamma_1 + \gamma_2 \hat{e}_t,$$

where $\hat{e}_t$ are ES forecasts

- Equivalently,

$$R_t = \gamma_1 + \gamma_2 \hat{e}_t + v_t^e,$$

where $ES_t(v_t^e|\mathcal{I}_{t-1}) = 0$ almost surely

- Standard Wald test for $H_0 : \gamma_1 = 0$ and $\gamma_2 = 1$

The same information set $\mathcal{I}_{t-1}$ is used to generate ES and in the backtest regression; so $ES_t(v_t^e|\mathcal{I}_{t-1}) = 0$ almost surely and eq 2 comed down to the first, so we can test ES forecasts validity .

–H0 is sufficient condition for correctly specified forecasts but not neessary; the necessary would be $\gamma_1 = (1 - \gamma_2)\hat{e}_t$, but for time-varying series it is unrealistic

The predictive performance of point forecasts is commonly measured in terms of the average realized score $1/n \sum_{t=1}^n S(X_t, Y_t)$ for some loss or scoring function S, using the orientation the smaller the better.

If a functional admits a strictly consistent score, it is called **elicitable**; ES is not elicitable, but VaR and ES are jointly elicitable (Fissler Ziegel 2016) (VaR is elicitable, based on the check function, it is a quantile).

Variance and ES are not elicitable, but bijections of an elicitable/identifiable functional is elicitable/identifiable. So the pair mean-variance or VaR-ES are elicitable and identifiable.

Backtesting ES

Bayer and Dimitriadis (2022)

Results of the empirical application

Rolling window	Refitting frequency					Re fitting frequency				
	5	21	62	125	250	5	21	62	125	250
	<u>GARCH- N</u>					<u>GJR-GARCH- N</u>				
250	0.96	0.96	0.95	0.95	0.92	0.96	0.96	0.95	0.92	0.90
500	0.94	0.94	0.93	0.91	0.87	0.93	0.93	0.94	0.94	0.91
1000	0.86	0.88	0.87	0.83	0.83	0.93	0.93	0.91	0.89	0.91
1500	0.85	0.88	0.86	0.87	0.87	0.93	0.93	0.93	0.93	0.91
2000	0.81	0.82	0.83	0.82	0.80	0.92	0.92	0.91	0.92	0.91
	<u>GARCH- t</u>					<u>GJR-GARCH- t</u>				
250	0.00	0.01	0.04	0.06	0.12	0.00	0.00	0.01	0.02	0.04
500	0.00	0.00	0.01	0.01	0.02	0.00	0.00	0.01	0.01	0.01
1000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.01
1500	0.00	0.01	0.00	0.01	0.01	0.00	0.01	0.00	0.00	0.01
2000	0.00	0.00	0.00	0.01	0.01	0.00	0.01	0.00	0.01	0.01

Notes: This table shows the rejection rates of the one-sided ESR backtest for ES forecasts stemming from the two GARCH-type models with Student's t and Gaussian residuals, different rolling window sizes, and model refitting lengths (in days). The rejection frequencies are averaged over the analyzed 200 most capitalized stocks of the S&P 500 index. The out-of-sample window covers the time from January 2010 to August 2019 resulting in a sample size of 2432 days.

Results from one-sided Intercept ESR test as this is the only one-sided and strict ES backtest in the literature.

refitting horizons - corresponding to weekly, monthly, quarterly, bi-yearly, and yearly updating of the models.

results show that both, the GARCH-N and GJR-GARCH-N are rejected for almost all the stocks (in more than 80% of the cases) uniformly over the different estimation sample sizes and refitting frequencies.

Independent of the sample length and refitting frequency, this supports the well-known finding that Gaussian residuals generally fail to capture the riskiness of financial assets, especially in the tails of the distribution.

In contrast, for the two GARCH specifications with Student-t distributed innovations, the rejection frequencies are considerably lower and for almost all choices of the refitting frequency and the estimation window length, they are below the nominal significance level of 5%.

Furthermore, refitting the models more frequently tends to slightly decrease the rejection frequency for the models with Student-t distributed innovations, however, it tends to increase the rejection frequency for the Gaussian models.

Overall, this implies that the refitting frequency is not a key factor in the model performance.

Increasing the size of the estimation window tends to decrease the rejection frequency, whereas the results stabilize for lengths above 1000 days.

Interestingly, employing the GJR-GARCH model, which accounts for a potential leverage effect in the volatility process, does not perform better than the standard GARCH model.

Overall, the results of this application which are diversified over 200 individual stocks, imply that using a fat-tailed residual distribution and an estimation window above 1000 days (roughly four years) suffices to obtain rejection rates uniformly below 1%.

Backtesting ES

- The reliability of any ES backtest procedure may be lower than that of VaR
 - With ES, we are testing whether the mean of returns on days when VaR is violated is the same as average ES on these days.
 - Harder (not impossible) to create formal tests to ascertain whether normalized ES equals one or not than the coverage tests developed above for VaR violations
- Hence, backtesting ES requires many more observations than backtesting VaR
- In instances where ES is obtained directly from VaR, and gives the same signal as VaR (i.e., when VaR is subadditive), it is better to simply use VaR
- Recent works bring a new perspective on this dilemma



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Du and Escanciano (2017)

- Use daily data for S&P 500, DAX, and Hang Seng Index (HS)
- Period : January 1, 1997 to June 30, 2009
- INS: January 1, 1997 to June 30, 2007
- OOS: July 1, 2007, to June 30, 2009

- crisis in oos
- the returns are leptokurtic and very volatile with big losses, especially during the crisis.
- Excess kurtosis is evident in all three indexes and dramatically so in HS because of the turmoil right after the return of Hong Kong to China in 1997.

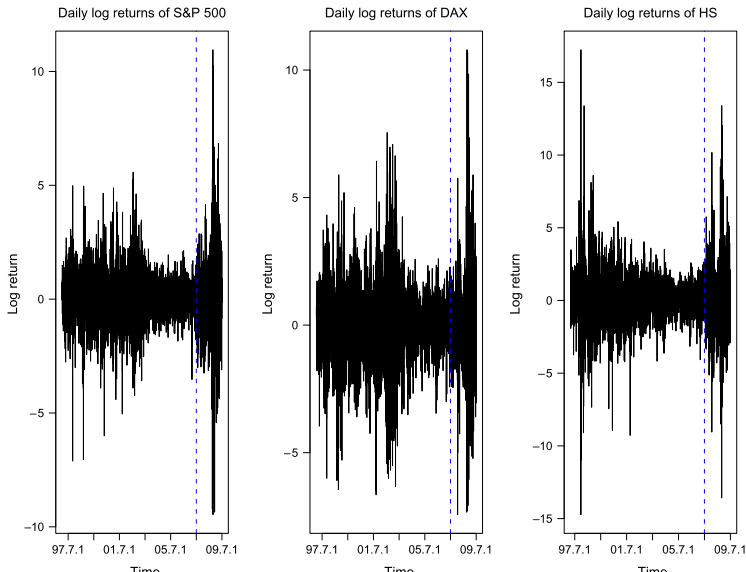
One can observe the volatility clustering feature of the data.

Moreover, the returns are more volatile during the financial crisis period, particularly for the S&P 500, which lost almost half of its value between July 2008 and the market bottom in March 2009.

Besides, there are some volatile periods for S&P 500 and DAX in 2001–2003 during the September 11 attack and stock market downturn; Hong Kong, on the other hand, experienced a volatile period in late 1997.

Du and Escanciano (2017)

Plots of the Daily Log Returns of S&P 500, DAX, and Hang Seng



Du and Escanciano (2017)

Descriptive Statistics for the Log Returns (%) of Three Indexes

	In-sample (1997–2007)			Out-of-sample (2007–2009)		
	S&P 500	DAX	HS	S&P 500	DAX	HS
No. of obs.	2,639	2,658	2,596	504	509	503
Mean	0.027	0.039	0.019	−0.098	−0.100	−0.034
Median	0.059	0.106	0.042	0.036	0.016	0.052
Std. dev.	1.131	1.577	1.677	2.218	2.045	2.762
Skewness	−0.089	−0.152	0.146	−0.065	0.351	0.167
Excess kurtosis	3.165	2.348	11.355	4.035	5.273	3.319
Maximum	5.574	7.553	17.250	10.960	10.800	13.410
10 percentile	−1.324	−2.132	−1.712	−2.478	−2.132	−3.238
5 percentile	−1.815	−2.567	−2.552	−3.506	−3.371	−4.381
1 percentile	−2.881	−4.547	−4.306	−6.310	−6.061	−7.305
Minimum	−7.113	−6.652	−14.73	−9.470	−7.433	−13.580

Du and Escanciano (2017)

- fit an AR(1)-GARCH(1,1) model with student's t innovations to the log returns
- estimate the parameters by CMLE using the in-sample data
- perform backtesting with the out of-sample data

Students - indicate the model and the risk measures

- $R_t = c_0 R_{t-1} + \varepsilon_t$, $\varepsilon_t = \sigma z_t$, $\sigma^2 = \omega_0 + \alpha_0 \sigma_{t-1}^2 \varepsilon_{t-1} + \beta + 0 \sigma_{t-1}^2$
- $VaR_t(\alpha) = -c_0 R_{t-1} - \sigma G^{-1}(\alpha, \nu)$
- $ES_t(\alpha) = -c_0 R_{t-1} - \sigma m(\alpha)$, with $m(\alpha) = \mathbb{E}[\varepsilon_t | \varepsilon_t \leq G^{-1}(\alpha, \nu)]$

Du and Escanciano (2017)

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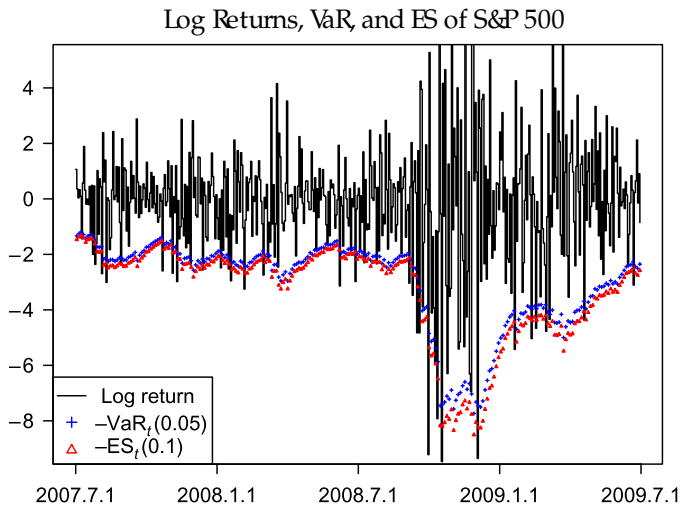
Du and Escanciano (2017)

CML Estimates

	S&P 500	DAX	Hang Seng
c_0	-0.027	0.004	0.034
ω_0	0.007	0.016	0.010
α_0	0.059	0.088	0.058
β_0	0.937	0.910	0.948
v	9	10	4
$G_v^{-1}(0.05)$	-1.617	-1.621	-1.507
$G_v^{-1}(0.01)$	-2.488	-2.472	-2.649
$m(0.1)$	-1.781	-1.779	-1.767
$m(0.025)$	-2.544	-2.521	-2.824

reports the CMLE estimates, including estimates for the terms $G^{-1}(\alpha, \nu)$ and $m(\alpha)$ the levels of α considered.¹² We observe a similar high level of volatility persistence for the three indexes. Hang Seng has the smallest student's t degree of freedom ν of the innovation distribution, and hence a fatter tail, in agreement with the high kurtosis of Hang Seng observed in the first Table .

Du and Escanciano (2017)



Figures 2–4 plot the estimated VaRt (0.05) and ESt (0.1) for the three series, respectively. One can see that -ESt (0.1) is smaller than -VaRt (0.05).

When Rt falls below -VaRt (0.05), -ESt (0.1) is closer to the true Rt compared with -VaRt (0.05). Take September 15, 2008, for example, when Lehman Brothers filed bankruptcy. S&P 500 fell by 4.83% on that day, and the estimated VaRt (0.05) is 1.82%, whereas ESt (0.1) is 2.65%, closer to the actual loss.

There are 41 cases out of 504 observations over the period from July 1, 2007, to June 30, 2009, where S&P 500 returns fall below their -VaRt (0.05).

The average loss of S&P 500 for those cases is 3.82%, and the average of VaRt (0.05) is 2.79%, whereas that of ESt (0.1) is 3.07%.

There are 11 cases where S&P 500 returns fall below their -VaRt (0.01). The average loss of S&P 500 for those cases is 3.76%, and the average of VaRt (0.01) is 3.13%, whereas that of ESt (0.025) is 3.20%.

Therefore, ES better describes the extreme losses than VaR. The results for DAX and HS tell similar stories.

Du and Escanciano (2017)

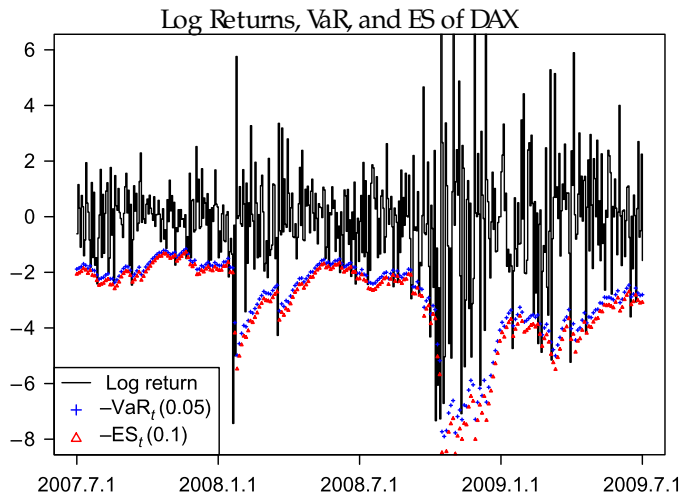
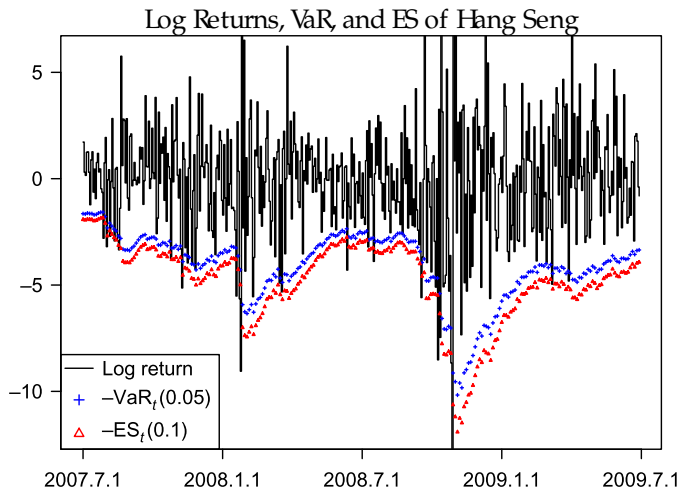


Table 12 reports the number of violations $V = \sum_t 1^n \hat{\delta}(\alpha)$, cumulative violations $CV = \sum_t 1^n \hat{C}V(\alpha)$, and the expected value of violations ($n\alpha$) for the three indexes in the precrisis and crisis periods. The precrisis period is here defined as the end of insample period with the same number of observations as the out-of-sample crisis period. Comparing V and CV before and after the crisis, one can see a significant increase of risk in the crisis in general. One exception is $V(0.01)$ of DAX, which actually drops in the crisis, whereas $CV(0.025)$ of DAX does increase in the crisis. If we take a further look, although the number of violations $V(0.01)$ drops, the losses are much larger in the crisis period than the precrisis period. This explains the increase of $CV(0.025)$ and the decrease of $V(0.01)$ in the crisis period. Table 12 shows significant discrepancies between violations and cumulative violations at the coverage level suggested by the Basel committee ($\alpha = 0.01$ for VaR).

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Descriptive Analysis of Violations: Precrisis vs. Crisis

	Precrisis (2005-2007)			Crisis (2007-2009)		
	S&P 500	DAX	Hang Seng	S&P 500	DAX	Hang Seng
$V(0.05)$	20	20	24	41	35	29
$CV(0.1)$	20.309	22.434	24.714	40.026	34.862	30.612
$n \times 0.05$	25.2	25.45	25.15	25.2	25.45	25.15
$V(0.01)$	5	8	2	11	5	5
$CV(0.025)$	6.110	6.360	4.063	13.702	9.101	6.145
$n \times 0.01$	5.04	5.09	5.03	5.04	5.09	5.03

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Cumulative Hits (Violations) of S&P 500, DAX, and Hang Seng

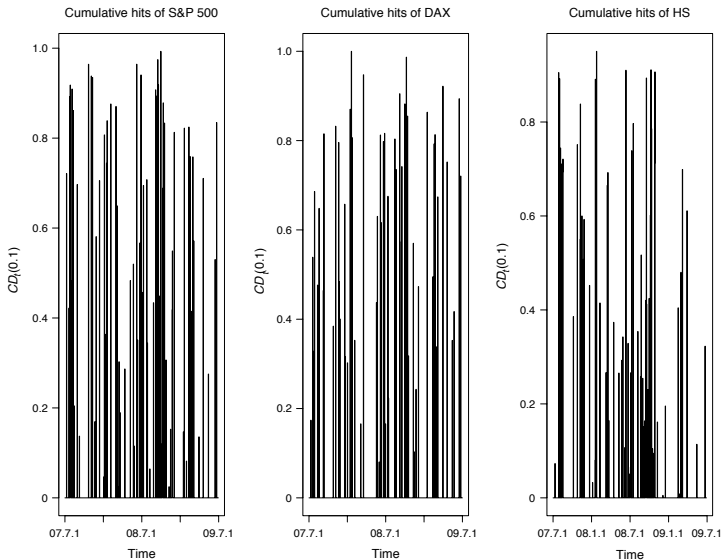


Figure 5 plots the cumulative violations $\{CV_t(0.1)\}$ of the three indexes in the out-of-sample crisis period. We observe large values of $\hat{CV}(0.1)$, which indicates a large loss on that day. As we can see there are more such cases for S&P 500 than DAX and Hang Seng. For the three indexes, there is substantial clustering of cumulative violations, which suggests deviations from the MDS hypothesis implied by an appropriate ES forecast. To formally assess this hypothesis, we apply our conditional backtest.

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 p -Values for Backtesting ES and VaR

	ES (0.025)	VaR (0.01)	ES (0.1)	VaR (0.05)
S&P 500				
U	0.011	0.070	0.004	0.010
$C(5)$	0.007	0.270	0.009	0.052
MU	0.019	0.073	0.006	0.013
$MC(5)$	0.017	0.271	0.010	0.053
DAX				
U	0.224	0.968	0.045	0.095
$C(5)$	0.002	0.998	0.091	0.768
MU	0.253	0.968	0.052	0.102
$MC(5)$	0.015	0.998	0.095	0.769
Hang Seng				
U	0.939	0.989	0.194	0.462
$C(5)$	0.002	0.998	0.002	0.002
MU	0.945	0.990	0.310	0.509
$MC(5)$	0.003	0.998	0.004	0.002

Note. U denotes the basic unconditional backtest; MU the robust unconditional backtest; $C(5)$ the basic conditional backtest with $m = 5$; $MC(5)$ the robust conditional backtest.

Table 13 reports the p-values of the unconditional tests U_{ES} , U_{VaR} , MU_{ES} , MU_{VaR} and the conditional tests $CES(5)$, $CVaR(5)$; $MCES(5)$, $MCVaR(5)$ for the three indexes, respectively. Our conditional backtests for ES, $CES(5)$, and $MCES(5)$, generally reject the null model, whereas $CVaR(5)$ and $MCVaR(5)$ do not. Figures 6 and 7 plot the sample autocorrelations of $\hat{C}Vt(0.025)$ and $\hat{\delta}t(0.01)$, respectively, from which we can also clearly see that the model for $ES(0.025)$ is rejected at 5% level, whereas the model for $VaR(0.01)$ is not.

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Sample Autocorrelations of Cumulative Hits of S&P 500, DAX, and Hang Seng

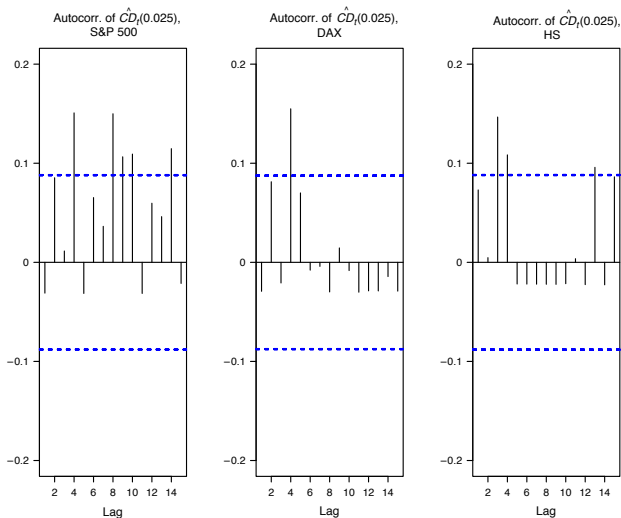
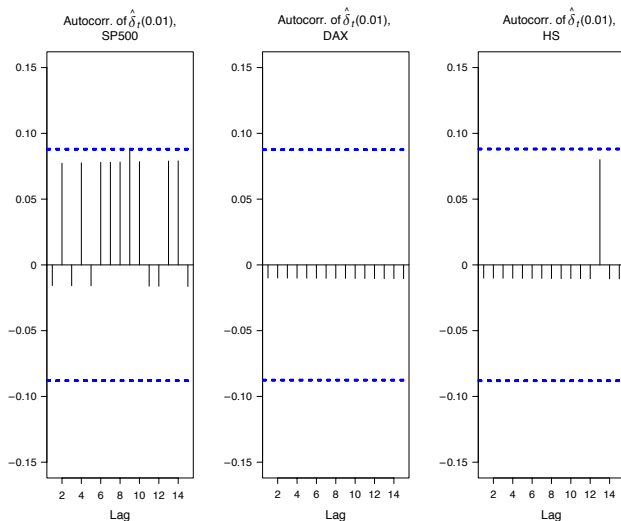


Figure 7 shows insignificant autocorrelations of $\{\hat{\delta}t(0.01)\}$ for all three series. The autocorrelations of $\hat{\delta}t(0.01)$ of DAX and Hang Seng are actually very close to 0 for the first twelve lags, as there are only five R_t 's falling below $-VaR_t(0.01)$ for those two indexes. The corresponding number for S&P 500 is eleven. The crisis originated and had a bigger impact in the United States, which brought about more extreme losses in the stock market in the United States than in Germany and Hong Kong. This also may explain why the unconditional test UVaR in Table 13 has a small p-value for S&P 500, and a big p-value for the other two indexes. The cumulative violations $\{\hat{C}Vt(0.01)\}$, on the other hand, have significant autocorrelations for all three series. The number of extreme losses may not be big, but the average losses can be big and highly correlated. The cumulative violations series $\{\hat{\delta}t(0.01)\}$ take both of those two pieces of information into account. On the contrary, conditional VaR backtests only look at clustering of tail events, and not to their magnitude. Therefore, as reported in Table 13, our test based on $\{\hat{\delta}t(0.01)\}$ is able to better detect the problems of one of the most commonly used risk models during the 2008 financial crisis.

In summary, based on VaR backtesting, one cannot find unambiguous empirical evidence against the AR-GARCH model with student's t innovations at the regulatory coverage level, and hence adjust the way the reserved capital is calculated during the financial crisis period. Instead, if one uses ES as the risk measure, our proposed backtesting procedure clearly rejects this model. Our empirical analysis here confirms that the theoretical advantages of ES over VaR documented in Artzner et al. (1997, 1999) also have empirical manifestations in the context of backtesting market risk. We have provided in this paper a set of tools based on cumulative violations that help assess not only the likelihood of financial losses but also the size of such losses.

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Sample Autocorrelations of Hits of S&P 500, DAX, and Hang Seng



Despite the substantial theoretical evidence documenting the superiority of ES over VaR as a measure of risk, only recently ES has been embraced by financial institutions and regulators as an alternative to VaR for financial risk management. Arguably, one of the major obstacles in this transition has been the unavailability of simple tools for the evaluation of ES forecasts (backtests).

In this paper, we have introduced cumulative violations as the building blocks for constructing unconditional and conditional backtests for ES, much like violations are the building blocks for the most commonly used backtests for VaR. Unlike violations, cumulative violations contain information on the tail risk and, therefore, provide a more complete description of the risk involved.

We have proposed basic unconditional and conditional backtests as well as modified backtests that are robust to the presence of estimation risk. The unconditional backtests check for the mean of cumulative violations with a t-test.

The proposed conditional backtests are Portmanteau tests applied to estimated cumulative violations. We also recommend to complement the information provided by formal tests with graphical tools such as the plot of cumulative violations and autocorrelograms of cumulative violations.

Our conditional backtest involves two choices that practitioners need to make: the coverage level α and the number of autocorrelations considered m . We have suggested choices for α such as 0.1, 0.05 and 0.025 for ES.

Smaller values are not recommended, as they would require very large out-of-sample sizes to achieve a satisfactory approximation of the finite sample distribution by the asymptotic distribution of tests. Regarding the choice of the number of correlations, we have suggested to apply the test with $m=5$.

A sensible alternative, however, is to consider a data-driven choice of m similar to that proposed in Escanciano and Lobato (2009b). This combined procedure has been shown to deliver simple and reliable inferences in other contexts, and it can be certainly used here to provide a fully data-driven backtests for ES at a small computational price.